

Membrane mimetic thermal proteome profiling (MM-TPP) towards mapping membrane protein-ligand dynamic interactions

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This **valuable** study introduces the peptidisc-TPP approach as a promising solution to challenges in membrane proteomics, enabling thermal proteome profiling in a detergent-free system. The concept is innovative and holds significant potential, and the demonstration of its utility and validation is **solid**. The method presents a strong foundation for broader applications in identifying physiologically and pharmacologically relevant membrane protein-ligand interactions.

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Abstract

Integral membrane proteins (IMPs) are central targets for small-molecule therapeutics, yet robust, unbiased, and detergent-free approaches to assess their on- and off-target interactions remain limited. Previously, we introduced the Peptidisc membrane mimetic (MM) for water-soluble stabilization of the membrane proteome and interactome (Carlson et al., eLife, 2019). In this work, we combine the Peptidisc with thermal proteome profiling (TPP) to establish membrane mimetic thermal proteome profiling (MM-TPP), a method that enables proteome-wide mapping of membrane protein–ligand interactions. Using a membrane protein library derived from mouse liver tissue, we detected the specific effects of ATP and orthovanadate on the thermal stability of ATP-binding cassette (ABC) transporters, as well as stability shifts driven by the hydrotropic effect of ATP and its by-products on G protein–coupled receptors (GPCRs). In contrast, detergent-based TPP (DB-TPP) with ATP–VO₄ failed to yield specific enrichment of ATP-binding proteins, underscoring the unique capacity of MM-TPP. To further validate the approach, we demonstrated the ability of MM-TPP to detect specific ligand-induced stabilization of cognate targets, exemplified by the selective thermal stabilization of the P2RY12 receptor by 2-methylthio-ADP. Together, these findings position MM-TPP as a robust platform for uncovering both on- and off-target effects of small molecules, providing insights into the druggable membrane proteome and its stability in consequence of changing dynamic ligands.

Introduction

Integral membrane proteins (IMPs) are essential to numerous cellular functions, including signal transduction, cell-cell recognition, and the transport of nutrients and ions across cellular membranes (Boulos et al., 2023 [↗](#)). Although comprising only 20 to 30% of the human genome, they represent nearly two-thirds of druggable targets due to their significant impact on cellular physiology and their exposure to the cell surface. This prevalence highlights the critical nature of IMPs in pharmacology (Bakheet and Doig, 2009 [↗](#); Fagerberg et al., 2010 [↗](#); Helbig et al., 2010 [↗](#); Santos et al., 2017 [↗](#)). However, studying the interactions between small-molecule drugs and membrane proteins poses unique challenges, primarily due to the low abundance and hydrophobic characteristics of these proteins, which complicate their characterization and analysis.

Proteomics-based technologies have become powerful tools for identifying protein–ligand interactions (Aebersold and Mann, 2016 [↗](#); Li et al., 2016 [↗](#)). Among these, affinity purification mass spectrometry (AP-MS) uses chemical compounds covalently attached to an affinity matrix to capture target proteins, allowing for detailed characterization of their interactions (Dunham et al., 2012 [↗](#); Kawatani and Osada, 2014 [↗](#)). Another approach, drug affinity responsive target stability (DARTS), leverages ligand-induced conformational changes detected through limited proteolysis, offering insights into the dynamic nature of protein–ligand interactions (Lomenick et al., 2009 [↗](#); Pai et al., 2015 [↗](#)). In addition to AP-MS and DARTS, thermal proteome profiling (TPP) has emerged as a complementary approach for studying protein–ligand interactions (Lambos et al., 2025 [↗](#)). Derived from the cellular thermal shift assay, TPP enables the detection of ligand-induced conformational changes by assessing protein stability under heat stress (Molina et al., 2013 [↗](#); Savitski et al., 2014 [↗](#)). Its relatively straightforward implementation has made it a popular choice in drug screening campaigns. However, TPP typically relies on cytosolic extracts and often overlooks pharmaceutically relevant transmembrane proteins (Franken et al., 2015 [↗](#); Mateus et al., 2020 [↗](#); Lambos et al., 2025 [↗](#)).

To expand these techniques to membrane proteome coverage, detergent-based solubilization has been widely used (Huber et al., 2015 [↗](#); Kalxdorf et al., 2021 [↗](#); Reinhard et al., 2015 [↗](#)). However, even mild detergents can disrupt native protein structures or drug associations, leading to artifacts that complicate drug target identification (Berlin et al., 2023 [↗](#); Yang et al., 2014 [↗](#); Ye et al., 2023 [↗](#)). While crucial for solubilizing membrane proteins and enabling structural and functional studies (O'Malley et al., 2011 [↗](#); Allison et al., 2015 [↗](#); Beckner et al., 2020 [↗](#)), detergents remain as we show here largely incompatible with workflows like TPP due to mass spectrometry (MS) limitations (Brough et al., 2024 [↗](#); Antony et al., 2025 [↗](#)).

To overcome the limitations associated with detergent use, a variety of MMs have been developed to maintain membrane proteins in a water-soluble, native-like state (Denisov and Sligar, 2016 [↗](#); Dörr et al., 2014 [↗](#); Young, 2023 [↗](#)). Among these, the Peptidisc has emerged as a versatile, self-assembling scaffold characterized by its “one-size-fits-all” property, capable of stabilizing IMPs of diverse sizes and topologies (Carlson et al., 2018 [↗](#)). This adaptability has enabled the comprehensive isolation of membrane proteomes from both bacterial and mammalian sources, yielding so-called Peptidisc libraries that are directly compatible with downstream MS workflows (Antony et al., 2024 [↗](#); Carlson et al., 2019 [↗](#); Zhao et al., 2023 [↗](#)). In addition to MS compatibility, these libraries also capture and stabilize IMPs in their functional states while preserving their interactomes and lipid allosteric modulators, thus enabling a more comprehensive analysis of membrane protein dynamics and ligand interactions (Angiulli et al., 2020 [↗](#); Carlson et al., 2019 [↗](#); Jandu et al., 2024 [↗](#); Urner et al., 2022 [↗](#)).

Building on this foundation, we thought to integrate the Peptidisc into the TPP workflow, introducing a straightforward and detergent-free approach termed membrane mimetic TPP (MM-TPP). Here we apply the MM-TPP workflow to bacterial and mouse liver membrane proteomes, demonstrating its ability to detect ATP-binding proteins, including ABC transporters, G protein-coupled receptors (GPCRs), and multi-subunit complexes. Notably, this approach is sensitive enough to capture interactions mediated by ATP by-products, offering critical insights into both system-wide and ligand-specific dynamics—factors that are essential to consider in small-molecule drug development.

Results and Discussion

The MM-TPP workflow is illustrated in **Figure 1**. Initially, the detergent-solubilized membrane fraction is reconstituted into Peptidisc libraries, as described in previous studies (Antony et al., 2024; Carlson et al., 2019; Zhao et al., 2023). The library is subsequently divided into two aliquots: one exposed to the ligand of interest (treatment), and the other treated with ddH₂O (control). Afterward, the samples are heated for three minutes to facilitate protein denaturation and precipitation. The soluble fraction is then isolated via ultracentrifugation and analyzed by liquid chromatography–tandem mass spectrometry (LC-MS/MS). Proteins exhibiting significant stabilization or destabilization are identified using the methodology outlined by Zhang et al. (2020), in which proteins that meet defined fold-difference thresholds between triplicate treatment and control groups are considered highly probable ligand binders.

As a validation step for the MM-TPP approach, we determined the thermal stability of the purified and Peptidisc-reconstituted bacterial ABC transporter MsbA in the presence of ATP and vanadate (VO₄). Vanadate is a potent inhibitor of the ABC transporter family and, upon binding with ATP, stabilizes the transporter in a specific conformation (Lyu et al., 2022). Consistent with this mechanism, SDS-PAGE analysis revealed increased thermal stability of MsbA in the presence of these ligands (**Figure 2A** and **Supplementary Figure 1**). To extend this observation to a more native context, we further assessed the effect of ATP-VO₄ on a Peptidisc library derived from wild-type *E. coli*, where MsbA is present at endogenous expression levels (**Figure 2B**). In the control sample, MsbA peptide abundance progressively decreased at 51, 56, and 61 °C relative to the bulk proteome. In contrast, in the presence of ATP-VO₄, MsbA was significantly stabilized at 51 °C and above (**Figure 2C**), demonstrating that our approach can detect thermal stabilization even in proteins expressed at low endogenous levels. Notably, at 61 °C, the volcano plot revealed that MsbA stabilization coincided with that of two other integral membrane proteins (IMPs), FtsK and LolC (**Figure 2D**). FtsK, one of the longest membrane proteins in *E. coli* (1329 amino acids), belongs to the AAA ATPase family, while LolC is the membrane transport domain of the soluble ABC subunit LolD (Sharma et al., 2021). In contrast, DgkA, which is a small diacylglycerol kinase comprising 122 amino acids, exhibited destabilization under the same conditions (**Figure 2D**) (Li et al., 2015; Zheng and Jia, 2013). Collectively, these results validate MM-TPP as a robust and straightforward method for identifying ATP binders within complex mixtures of soluble and integral membrane proteins.

We next applied the MM-TPP workflow to mouse liver tissue, where drug screening for cell surface IMPs is particularly relevant. As expected, the total number of proteins identified decreased following heat treatment (**Figure 3A**). Notably, in this detergent-free environment, this reduction affected soluble and membrane proteins equally, with the proportion of IMPs remaining stable across temperatures—approximately 48% (**Supplemental Table 1**). We then performed MM-TPP in the presence of ATP-VO₄ (**Figure 3B**). Consistent with results from the *E. coli* library, most proteins exhibited increased thermal stability upon ATP-VO₄ treatment, as indicated by the rightward shift of data points in the plot. Recent studies suggest that ATP can function as a natural hydrotrope, influencing the global stability of protein populations by interacting with thermally sensitive regions (Ou et al., 2021; Patel et al., 2017). Important to this study and as expected,

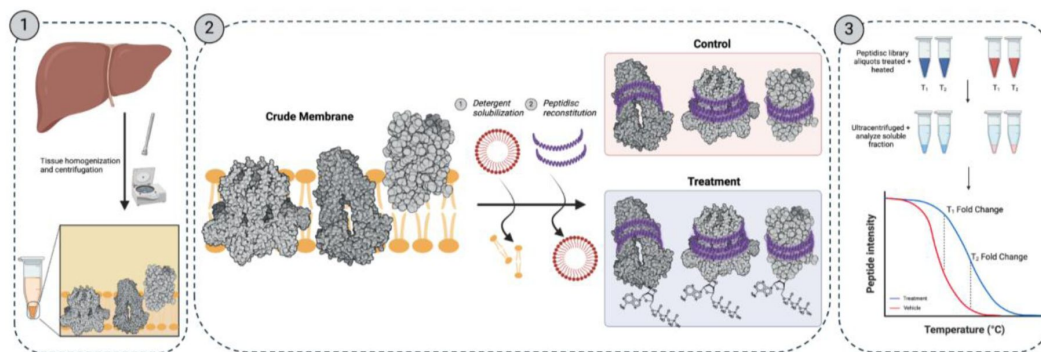


Figure 1

The Membrane Mimetic-Thermal Proteome Profiling (MM-TPP) experimental workflow.

1) Crude membranes are prepared from the liver organ. 2) Integral membrane proteins (IMPs) are solubilized with detergent and reconstituted in the Peptidisc library. The water-soluble library is exposed to the ligand of interest (treatment) or the corresponding vehicle (control). 3) Protein samples are heated at specific temperatures to induce precipitation, followed by ultracentrifugation. The soluble fraction is analyzed by mass spectrometry to detect changes in protein abundances between the treatment and control samples.

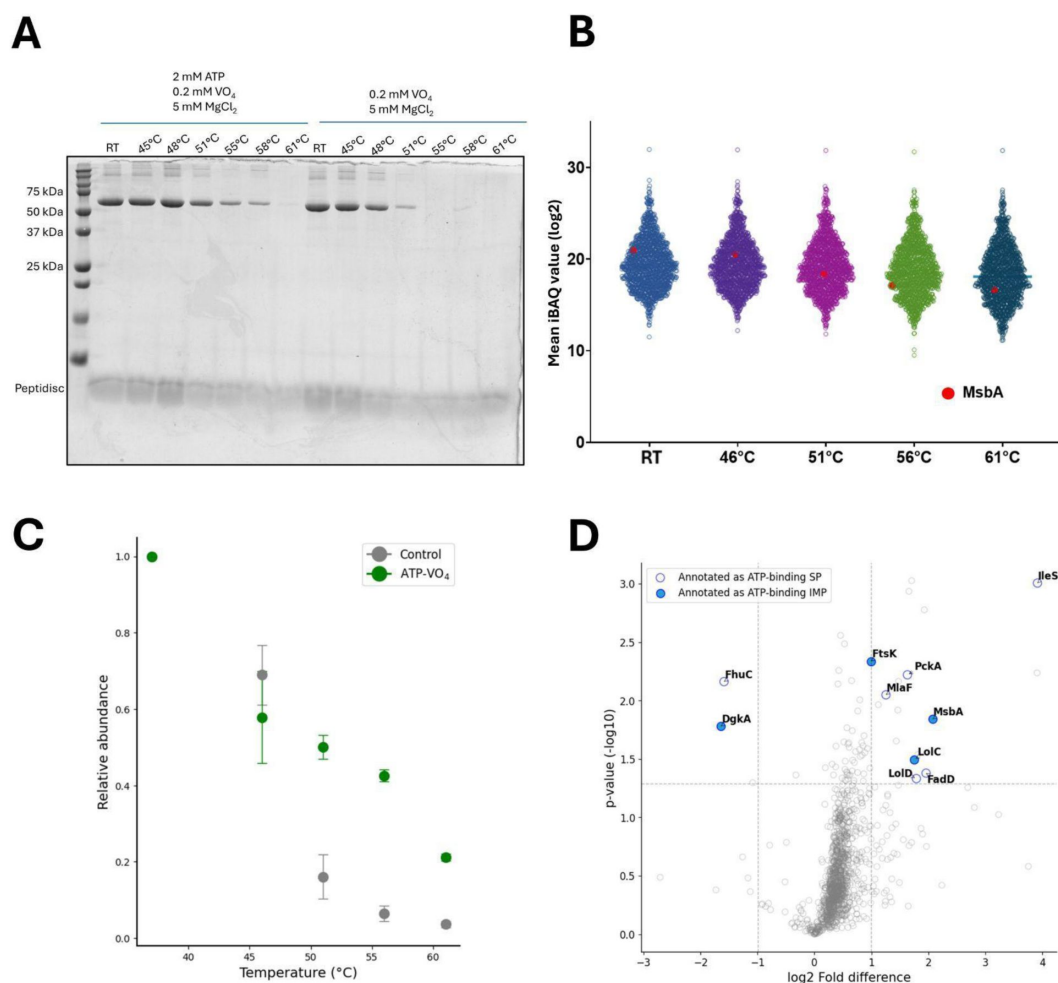


Figure 2

MM-TPP of integral membrane proteins (IMPs) prepared from *E. coli*.

(A) Stability of purified MsbA in Peptidisc in the presence of the indicated ligands. Samples are heat treated and centrifuged before analysis by 12% SDS-PAGE and Coomassie blue staining. (B) Grouped scatterplot representation of mean $\log_2$ -transformed IBAQ value obtained for all identified proteins in the *E. coli* library at the indicated temperatures. The location of MsbA on the plot is shown as a red dot. The mean value is obtained from three replicates of the temperature exposure assay ($n = 3$). (C) Relative abundance of MsbA based on the LFQ peptide intensities obtained across temperatures in the presence of ATP + orthovanadate (ATP- VO_4 ; orange) compared to a control sample (gray). Data is a mean $\pm$ standard deviation from triplicate assays ($n = 3$). (D) Volcano plot analysis of stabilized and destabilized proteins following ATP- VO_4 exposure at 61°C based on $\log_2$ -transformed LFQ peptide intensities. A $\log_2$ fold difference significance cut-off of +1 and -1 with a $-\log_{10}$ p-value cut-off of $p > 1.3$ is applied. Hollow blue dots indicate annotated ATP-binding soluble proteins (SP) and solid blue dots indicate ATP-binding integral membrane proteins (IMPs). Data represent the mean from three replicates ($n = 3$).

GO-term analysis of significantly stabilized IMPs revealed enrichment in functions related to nucleoside-phosphate binding and primary active transport (**Figure 3C**). Supporting this, thermal stability profiles across different temperatures confirmed a marked sensitivity of ABC transporters. Among the ten ABC transporters identified in the liver library, eight showed significant thermal stabilization in the presence of ATP-VO₄ at one or more temperature points (**Table 1** and **Figure 3D**). Besides ABC transporters, the BCS1L protein exhibited a remarkable ~30-fold increase in stability, which was further enhanced by AMP-PNP (**Figure 3E** and **Supplementary Figure 2**). This enhanced stabilization is consistent with recent structural studies demonstrating substantial conformational changes in the heptameric BCS1L complex upon binding of this non-hydrolyzable ATP analog (Pan et al., 2023; Tang et al., 2020; Zhan et al., 2024).

To further assess the effectiveness of MM-TPP, we directly compared its performance to a detergent-based TPP assay (DB-TPP) at 51 °C. In both workflows, crude membrane fractions from mouse liver were solubilized in 1% DDM to extract membrane proteins. However, while MM-TPP involves reconstitution of solubilized proteins into Peptidiscs before heat treatment, the DB-TPP workflow omits this step, keeping the proteins in detergent solution throughout. Following thermal treatment, DB-TPP also includes an additional detergent removal step for MS analysis. This side-by-side comparison revealed key differences in the TPP profiles. Overall, DB-TPP provided broader proteome coverage, including a greater number of soluble proteins identified (**Table 2**). In contrast, in the MM-TPP workflow, soluble proteins were actively depleted during Peptidisc reconstitution, resulting in an IMP-enriched proteome. Crucially, at 51 °C, 7.4% of all identified proteins and 6.4% of differentially stabilized proteins in the DB-TPP dataset were annotated as ATP-binding, indicating that the apparent differential stabilization is likely stochastic rather than driven by specific ligand binding (**Figure 4A**). By comparison, in MM-TPP, 9.3% of all identified proteins and 17% of differentially stabilized proteins were annotated as ATP-binding, demonstrating selective enrichment for ATP-VO₄-mediated stabilization events (**Figure 3B**).

Beyond demonstrating improved selectivity for ATP-VO₄-mediated stabilization, we next asked whether MM-TPP could also resolve more selective ligand–protein interactions. To test this, we applied the method using 2-methylthio-ADP (2-MeS-ADP), a well-characterized agonist of the P2RY12 receptor (Zhang et al., 2014). MM-TPP revealed clear and reproducible thermal stabilization of P2RY12, showing more than a sixfold increase in stability at both 51 °C and 57 °C (–log₁₀ p > 5.97; **Figure 4B** and **Figure S4**). Notably, no other proteins, including the structurally related but non-responsive P2RY6 receptor, exhibited comparable stabilization, underscoring MM-TPP's ability to detect highly specific ligand–receptor interactions amidst a complex membrane proteome.

Next to these highly specific receptor–ligand interactions, we also observed a broader spectrum of thermal stabilization events that lacked direct annotation to ATP binding. For instance, a substantial proportion of IMPs without known ATP-binding function were still significantly stabilized in the presence of ATP-VO₄. For example, among the 178 IMPs that passed the significance threshold, approximately 43% were annotated with GO terms related to nucleic acid/nucleotide binding, GTP binding, phosphate binding, and other related functions (**Figure 5A**). Some of these IMPs may achieve stabilization indirectly through interactions with ATP-binding partners, as illustrated by the LolC/LolD complex reported above (**Figure 2D**). Others may exhibit off-target stabilization effects, potentially due to ATP metabolites such as ADP, AMP, or inorganic phosphate, which are known to influence protein conformation and stability. This scenario is exemplified by the purinergic receptor P2RY6, a GPCR that responds preferentially to ADP rather than ATP (von Kügelgen and Hoffmann, 2016), which showed the highest differential thermal stabilization among all IMPs at 56 °C (**Supplementary File**). Similarly, P2RY12, a member of the same GPCR family, showed significant stabilization across 56, 60, and 64 °C (**Supplementary File**), consistent with its known preference for ADP (Entsie et al., 2023; Zhang et al., 2014). Supporting this interpretation, neither receptor exhibited thermal stabilization in the presence of

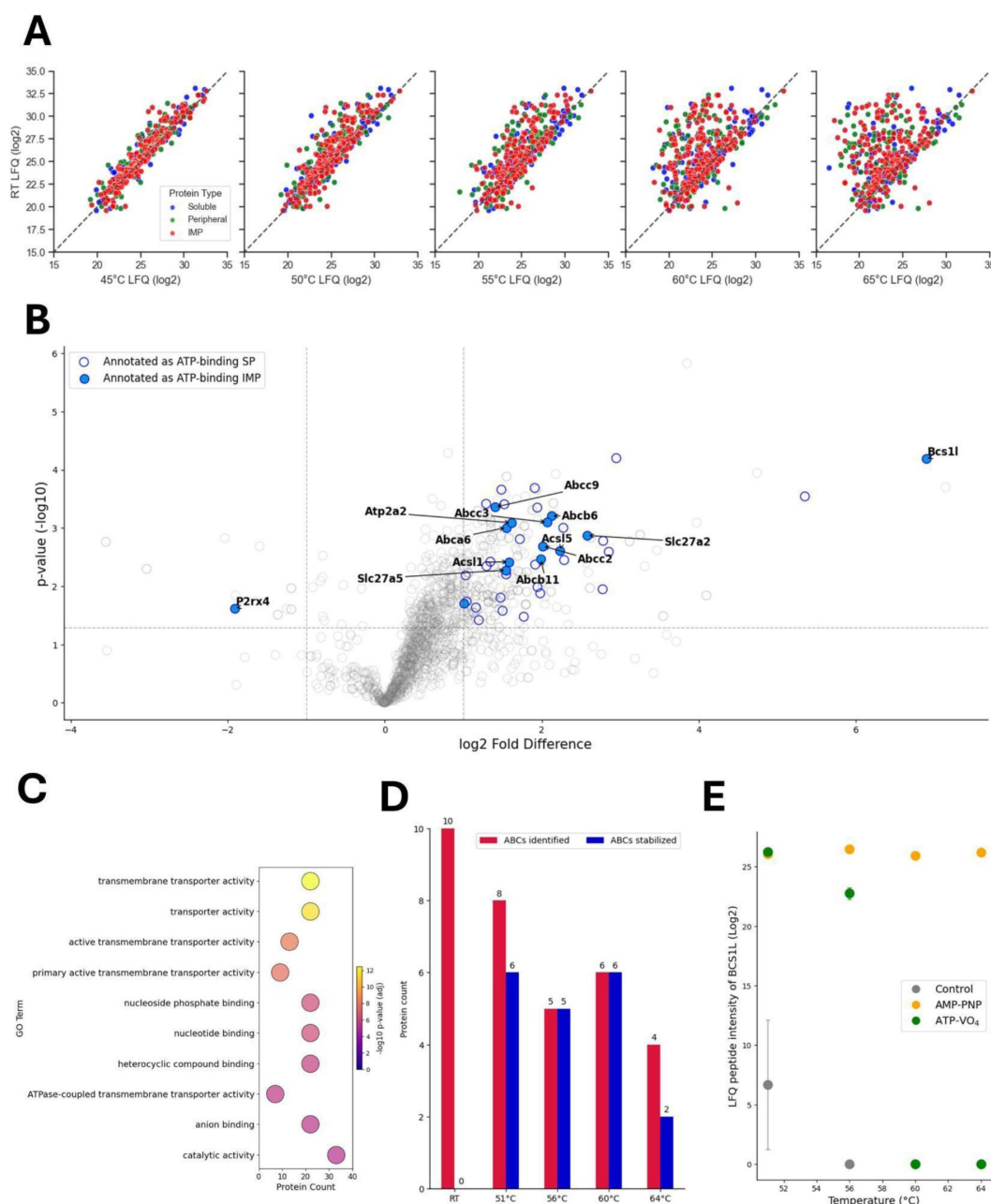


Figure 3

MM-TPP of integral membrane proteins (IMPs) prepared from the mouse liver tissue.

(A) Global protein intensities derived from log₂-transformed LFQ values of peptidisc-reconstituted liver extract. The plot displays soluble proteins (blue), membrane-associated (green), and IMPs (red) identified at the indicated temperatures compared to room temperature (RT). The dashed line is the identical value line. (B) Volcano plot of stabilized and destabilized proteins based on log₂-transformed LFQ values at 51 °C based on a fold difference cutoff of > 1 or < -1 and -log₁₀ p-value of > 1.3. The soluble proteins (SP) annotated as ATP-binding are represented as hollow blue circles, and the IMPs annotated as ATP-binding are represented as solid blue circles. The mean value is obtained from three replicates at the temperature exposure assay (n = 3). (C) GO-term enrichment analysis of molecular functions of stabilized IMPs identified in B. The presented top 10 significant terms are based on adjusted p-value (FDR = 5% after Benjamini-Hochberg correction) (D) Number of ATP-binding cassette (ABC) transporters identified and stabilized by ATP-VO₄ at the indicated temperatures. (E) Log₂ transformed LFQ Peptide intensities of BCS1L in the presence of ATP-VO₄ (green), AMP-PNP (orange), and vehicle control (gray) across the temperature range. Data is a mean ± standard deviation from three replicates (n = 3).

Uniprot-ID	Protein Name	Full protein name	Stabilized temperatures (°C)
Q8K441	ABCA6	ATP-binding cassette sub-family A member 6	51, 56, 60, 64
J3QNY6	ABCB11	ATP-binding cassette, sub-family B (MDR/TAP), member 11	51, 56, 60
Q9DC29	ABCB6	ATP-binding cassette sub-family B (MDR/TAP), member 6	51
Q8VI47	ABCC2	ATP-binding cassette sub-family C (CFTR/MRP), member 2	51, 56, 60
A0A0R4J015	ABCC3	ABC-type glutathione-S-conjugate transporter (CFTR/MRP)	51, 60
P70170	ABCC9	ATP-binding cassette sub-family C member 9 (Sulfonylurea receptor 2)	51, 56, 60
S4R2E1	ABCG2	ATP-binding cassette sub-family G member 2) (Urate exporter)	56, 60, 64
Q99PE8	ABCG5	ATP-binding cassette sub-family G member 5 (Sterolin-1)	51

Table 1

ATP-binding cassette transporters detected in the mouse liver Peptidisc library.

A minimum of two unique peptides was required to identify the protein at a given temperature. The exact number of unique peptides identified for each protein under each ligand and temperature condition is provided in **Supplementary Table 2** [2](#). Stabilization was defined using a $\log_2$ fold change > 1 between treatment and control samples, with a $-\log_{10} p > 1.3$, calculated from replicate samples ($n = 3$).

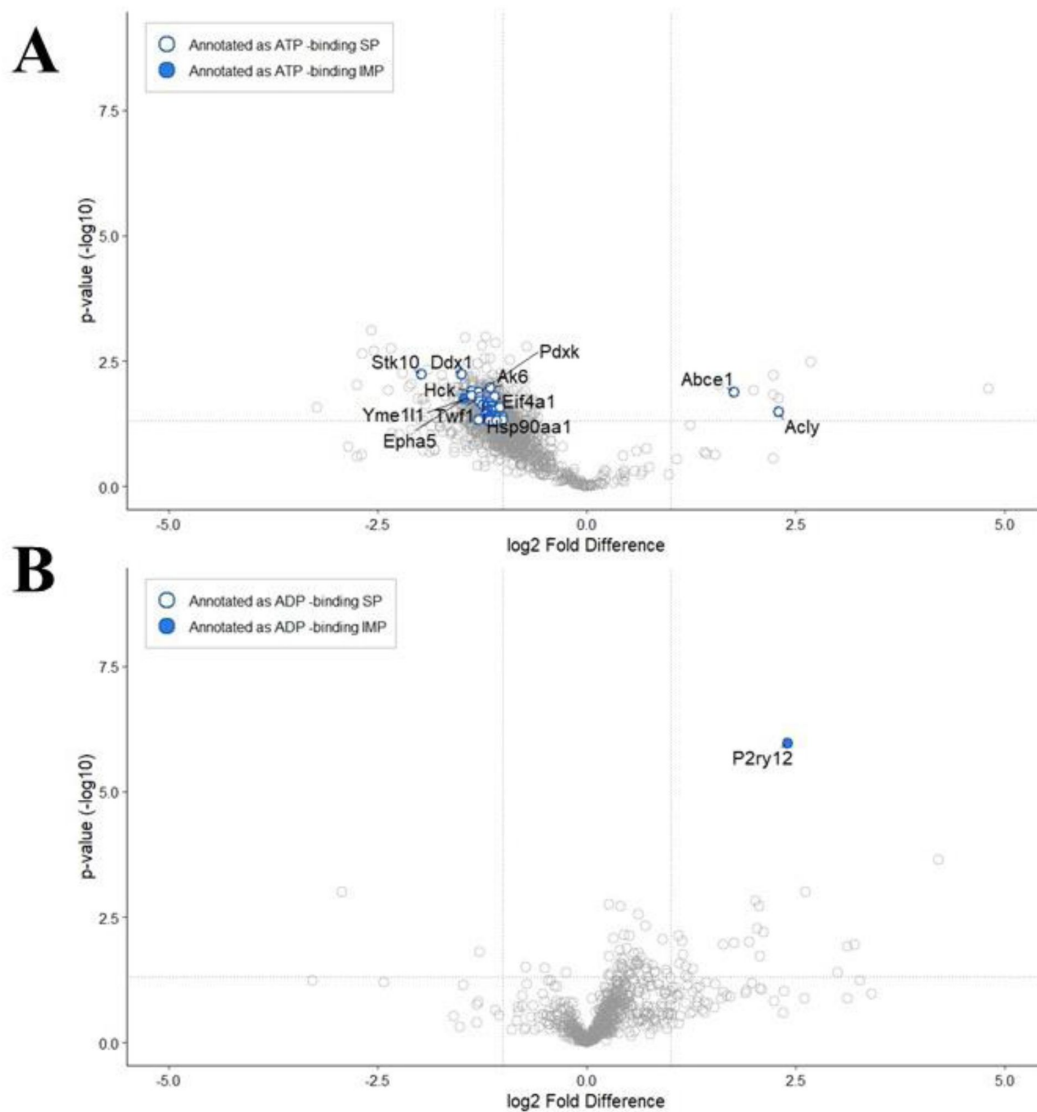


Figure 4

Protein Stabilization Profiles at 51 °C with ATP-VO₄ in DB-TPP and 2-MeS-ADP in MM-TPP.

(A) Volcano plot of stabilized and destabilized proteins at 51 °C in detergent-based TPP (DB-TPP) with ATP-VO₄. Log₂-transformed LFQ peptide values were used, with thresholds set at a fold-change > 1 or < -1 and a -log₁₀ p-value > 1.3. Soluble proteins (SP) annotated as ATP-binding are shown as hollow blue circles, and integral membrane proteins (IMPs) annotated as ADP-binding are shown as solid blue circles. Data represent the mean of three biological replicates (n = 3). (B) Volcano plot of stabilized and destabilized proteins at 51 °C in MM-TPP with 2-methylthio-ADP (2-MeS-ADP), displayed using the same thresholds and annotations as in panel A.

Temperature (°C)	MM-TPP				DB-TPP			
	Total Protein	IMPs	SPs	IMPs	Total Protein	IMPs	SPs	IMPs
			% ATP binders stabilized	% ATP binders stabilized			% ATP binders stabilized	% ATP binders stabilized
51	1380	419 (30%)	33.7%	37.9%	1862	428 (23%)	1.74%	0%
56	1179	369 (31%)	39.2%	45.6%	1839	428 (23%)	0%	0%
60	1090	344 (32%)	22.9%	48.4%	1621	391 (24%)	0%	0%
64	992	316 (32%)	47.0%	58.1%	1539	372 (24%)	0%	5.56%

Table 2

Comparison of ATP-Binding Protein Stabilization in MM-TPP and DB-TPP.

Reported values at each temperature represent the total number of proteins meeting the inclusion criteria (at least 2 unique peptides) across triplicate control and treatment conditions. As protein counts are determined post-analysis, they are reported as single values rather than as means with standard deviations. Each protein was identified based on at least two unique peptides (n = 3).

the non-hydrolyzable ATP analog AMP-PNP (**Figure 5B**). Conversely, the trimeric ATP-gated cation channel P2RX4 showed significant destabilization in response to ATP-VO₄ at 51 and 64 °C (**Figure 3B**) and AMP-PNP at 56 and 64 °C (**Figure 5C**), consistent with previous reports highlighting the unique ATP sensitivity of P2RX family members compared to the P2RY family (Carnero Corrales et al., 2021; Thompson et al., 2021).

Among all observed side-effects, the most striking stabilization was detected for the FAD-containing monoamine oxidase B (Mao-B) at 64 °C (**Supplementary Figure 3**). To explore the potential basis for this off-target effect, we used AlphaFold3 to model the binding of ATP and its derivatives within the FAD-binding pocket of Mao-B (Abramson et al., 2024). Interestingly, the best-fitting models were obtained for ADP and AMP (pLDDT > 90), while ATP yielded the lowest confidence score (**Figure 5D**). Although this computational result remains hypothetical, it offers a plausible structural rationale that warrants further experimental validation.

Finally, to validate the robustness of these findings, we assessed the reproducibility of MM-TPP by repeating the ATP-VO₄ experiment at 51 °C using data-independent acquisition (DIA) (**Supplementary Figure 5**), in contrast to the data-dependent acquisition (DDA) employed initially. The results were highly consistent with the previous dataset. Overall, 7.8% of all identified proteins were annotated as ATP-binding, with this proportion rising to 17% among proteins exhibiting a log₂ fold change greater than 0.5—mirroring the enrichment observed in the DDA-based analysis. Notably, BCS1L and SLC27A2 displayed strong thermal stabilization (log₂ fold change > 1), while several ABC transporters, including ABCB11, ABCG2, ABCD3, ABCG8, and ABCA1, showed intermediate stabilization (log₂ fold changes between 0.5 and 1). In line with earlier results, P2RX4 was again significantly destabilized (log₂ fold change < -1). These findings confirm that MM-TPP produces consistent and reproducible profiles of protein stabilization and destabilization across distinct MS acquisition strategies, further supporting its utility for ligand engagement studies in complex membrane proteomes.

Together, these results validate the integration of Peptidisc libraries into TPP for broad-scale detection of membrane protein–ligand interactions. MM-TPP not only enhances the recovery and stability of IMPs but also enables direct profiling of membrane proteomes from native tissues and organs. This preserves critical biological features, such as endogenous protein expression levels, native protein–protein interactions, post-translational modifications, and tissue-specific isoforms, that are often disrupted in traditional cell culture systems (Kwasnik et al., 2016; Perez-Perri et al., 2023).

Although the Peptidisc workflow does not bypass the need for initial detergent solubilization, it eliminates key downstream limitations associated with detergent-based methods. In particular, MM-TPP preserves ligand binding and is fully compatible with mass spectrometry, unlike many detergent-based workflows where residual detergent can interfere with both ligand interactions and MS sensitivity (Brough et al., 2024; Lambos et al., 2025). As such, MM-TPP is inherently complementary to detergent-based approaches while offering clear advantages for downstream proteomic analyses. To further expand its applicability, future studies could explore a broader range of nucleotide analogs, including antagonists such as TNP-ATP or ATPγS, and clinically relevant small molecules, such as the CFTR modulator VX-809. These applications would further underscore MM-TPP's pharmacological potential in profiling membrane protein function and drug interactions.

While challenges remain, such as maintaining physiological signaling contexts, integrating MMs like Peptidiscs early in drug discovery offers a powerful strategy. MM-TPP captures both direct target engagement and off-target effects from parent compounds or their metabolites, providing functional insights into ligand dynamics and specificity, which are important considerations in early-stage drug development and safety profiling.

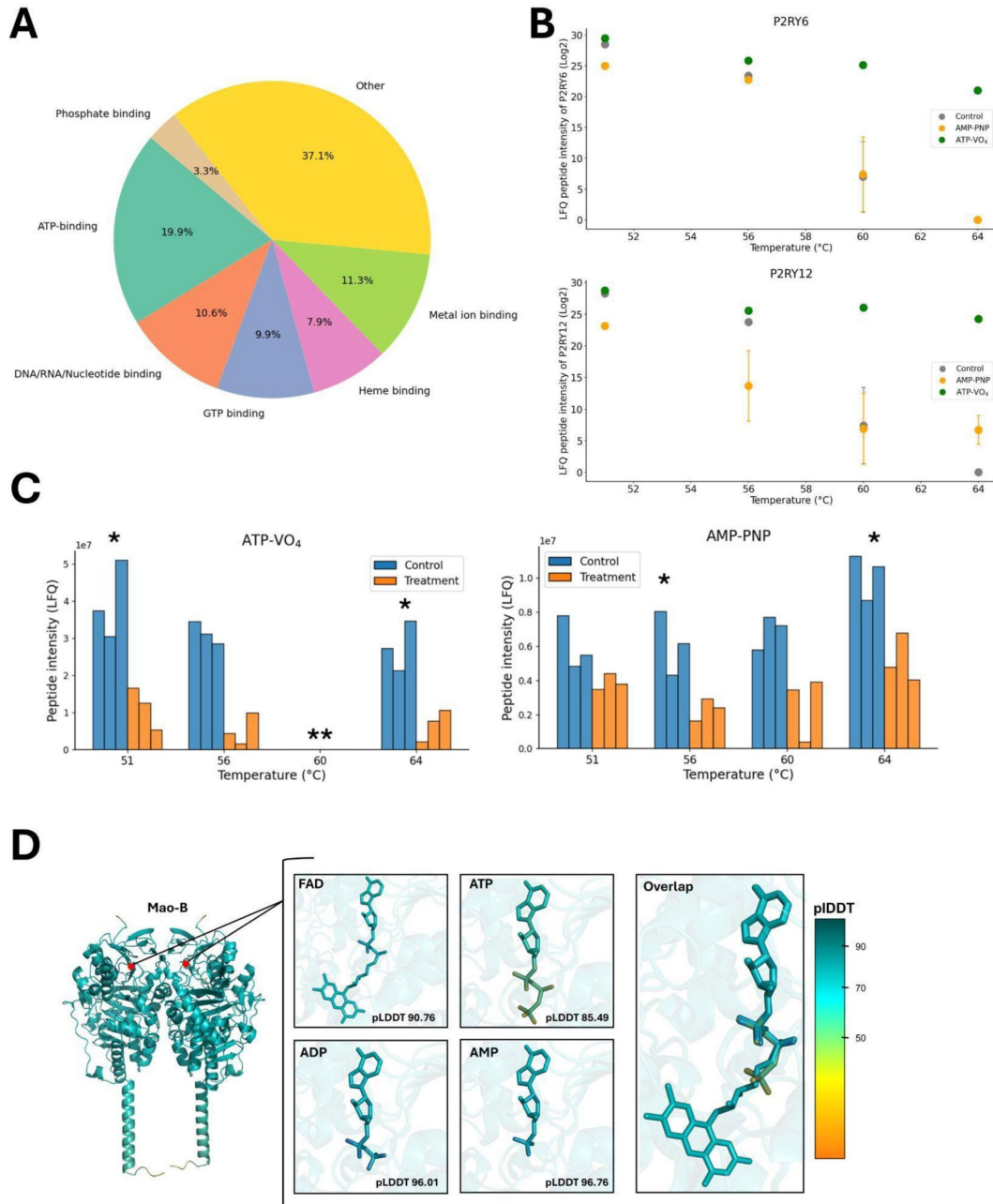


Figure 5

ATP and Analogue-Induced Off-Target Stabilization of Liver Membrane Proteins.

(A) GO-term analysis of molecular functions and distribution for all IMPs (n = 178) significantly stabilized with ATP-VO₄ at temperatures tested with the mouse liver library. (B) Log₂ transformed LFQ Peptide intensity variations of P2RY6 and P2RY12 over the temperature range with AMP-PNP (orange), ATP-VO₄ (green), or none (gray). Data is a mean ± standard deviation from three replicates (n = 3). (C) Relative LFQ Peptide intensity variations of P2RX4 at the indicated temperature in the presence of ATP-VO₄ (left panel) or AMP-PNP (right panel). Data from treatment samples (orange) and control samples (blue) is from triplicates (n = 3). * Represents p-value ≤ 0.05. **Protein not detected. (D) Structural model of homodimeric Mao-B with the predicted binding of FAD, ATP, ADP, and AMP ligands within the FAD binding pocket, indicated as red dots. Each ligand is presented individually in the FAD binding pocket or as an all-ligand overlap generated by AlphaFold3. The respective Predicted Local Distance Difference Test (pLDDT) score for each ligand is shown, with higher scores representing more favourable ligand fitting. The colour gradient represents a high pLDDT score as blue and a low pLDDT score as orange.

Methods

Preparation of *Escherichia coli* crude membranes

Wild-type *Escherichia coli* BL21 (DE3) was grown in 1 L of LB medium. After 3 hours, cells were harvested by low-speed centrifugation (6000g, 6 min) and resuspended in Buffer A (50 mM Tris-HCl pH 7.8, 100 mM NaCl, 10% Glycerol) supplemented with 1 mM phenylmethylsulfonyl fluoride (1 mM). Cells were lysed through a microfluidizer (Microfluidics; 3 passes at 15,000 psi at 4 °C). Unbroken cells and large aggregates were removed by low-speed centrifugation (6000g, 6 min). The crude membrane fraction was isolated by ultracentrifugation (100000g, 45 min, 4 °C, Beckman Coulter rotor Ti70). Membranes were resuspended in Buffer A at 5 mg/mL and stored at -80°C for later use.

Expression and purification of His-MsbA for thermal stability assay

Histidine-tagged MsbA (His-MsbA) was expressed, purified, and reconstituted in peptidase through the on-bead method as described in Angiulli et al. (2020) [with slight modifications](#) (Angiulli et al., 2020). Briefly, His-tagged MsbA was produced in *E. coli* BL21(DE3) at 37°C in 1 L of LB medium supplemented with 50 µg/mL kanamycin. The inducer IPTG (0.5 mM) was added during the exponential growth phase ($OD_{600nm} \sim 0.4$). After 3 hours, cells were harvested by low-speed centrifugation (6000g, 6 min) and resuspended in Buffer A (50 mM Tris-HCl pH 7.8, 100 mM NaCl, 10% Glycerol) supplemented with phenylmethylsulfonyl fluoride (1 mM). Cells were lysed through a microfluidizer (Microfluidics; 3 passes at 15,000 psi at 4°C). Unbroken cells and large aggregates were removed by low-speed centrifugation (6000g, 6 min). The crude membrane fraction containing MsbA was isolated by ultracentrifugation (100000g, 45 min, 4°C, Beckman Coulter rotor Ti70). About 2 mg of MsbA-enriched membranes were solubilized with 1% DDM (w/v) for 30 min at 4°C. The detergent solubilized material was ultracentrifuged (180000g, 15 min, 4°C, Beckman Coulter rotor TLA55) to pellet insoluble material. The supernatant was incubated with Ni-NTA (150 µL) resin for 45 min at 4°C on a tabletop rocker. The resin was sedimented through centrifugation (2000g, 1 min) and washed three times with 1.5 mL of Buffer A supplemented with 0.02% DDM and 30 mM imidazole. After removing the excess buffer, the beads were resuspended in Buffer A containing Peptidase peptide (1 mg/mL). The excess peptide was washed away with Buffer A. The protein was eluted with 600 mM imidazole in Buffer A and the concentration was determined through a Bradford reagent.

Thermal stability assay of His-MsbA with ATP-VO₄

Purified His-MsbA (0.5 mg/mL) in Buffer A supplemented with 5 mM MgCl₂ and 0.2 mM orthovanadate (VO₄) was separated into two equal volumes designated as treatment and control samples. To the treatment sample ATP-disodium-trihydrate (ATP) was added to a final concentration of 2 mM and an equal volume of ddH₂O was administered to the control. Samples were incubated at room temperature for 10 minutes prior to aliquoting 50 µL into separate 0.2 µL PCR tubes. These aliquots were subjected to temperatures ranging from 45 - 61°C for three minutes, followed by ultracentrifugation at 180000g for 15 minutes at 4°C to pellet the unfolded proteins. The supernatants were collected, and 10 µL from each sample was analyzed on a 12% SDS-PAGE.

Harvest of *Mus musculus* liver

The C57BL/6 mice were kept in specific pathogen-free conditions and received humane care in compliance with the Canadian Council of Animal Care guidelines, and the animal protocol was approved by the Animal Care Committee of the University of British Columbia. The mouse organs were obtained from female mice. The authors acknowledge that they did not consider the impact

of mouse sex at the time of the study design. The mice received standard chow. At the age of 12 weeks, the mice were sacrificed, and the liver was excised and placed in vials with ice-cold sterile phosphate-buffered saline (PBS) until processing.

***Mus musculus* liver processing and preparation of crude membranes**

The excised organs were washed several times in ice-cold PBS to remove the blood and then minced and homogenized in ice-cold hypotonic lysis buffer (10 mM Tris-HCl pH 7.4, 30 mM NaCl, and 1 mM EDTA, 1× cocktail protease inhibitor, and 1 mM PMSF) using a tight-fit metal douncer. All subsequent steps were performed at 4 °C or on ice. After adding 10 mM MgCl₂ and 50 µg/mL DNase, the suspension was further dounced and incubated for 10 min. The swollen tissue suspension was lysed using a French press (3 passes at 500 PSI). Unbroken cells and nucleus fraction were removed by low-speed centrifugation (1200g for 10 min). The supernatant was collected and centrifuged at a higher speed (5000g, 10 min) to remove the mitochondrial fraction. The crude membrane fraction was then collected by ultracentrifugation (110000g, 45 min) in a Beckman TLA110 rotor. This membrane pellet was resuspended in 100 µL of 50 mM Tris, pH 7.9, 100 mM NaCl, and 10% glycerol (TSG Buffer) and stored at -80 °C until use.

Preparation of Peptidisc membrane protein library

For Peptidisc library preparation, ~2 mg of crude membranes (either *Escherichia coli* or *Mus musculus*) was solubilized in 50 mM Tris, pH 7.9, 25 mM NaCl, and 1% DDM (Solubilization Buffer) for 30 min at 4 °C with gentle shaking. The insoluble material was pelleted by ultracentrifugation (180000g, 15 min). The detergent extract (500 µL) was then reconstituted by mixing it with a 3-fold excess (w/w) Peptidisc peptide for 15 min at 4 °C. The sample was rapidly diluted to 15 mL in 50 mM Tris, pH 7.9 and 25 mM NaCl (Buffer A) over a 100 kDa cut-off centrifugal filter. The sample was concentrated (3000g, 10 min) to ~200 µL. This process was repeated for a total of three rounds of dilution and concentration to deplete DDM to an approximate concentration of 0.008% to complete Peptidisc reconstitution. The Peptidisc library was immediately used for downstream thermal proteome profiling.

MM-TPP profiling on membrane proteomes

Peptidisc membrane protein library was split into two equal aliquots representing the control and treatment samples. Both the treatment and control groups were supplemented with 5 mM MgCl₂ and 0.2 mM VO₄. Treatment samples were then exposed to a final concentration of 2 mM ATP, while the control received an equal volume of ddH₂O. For experiments with AMP-PNP, 2 mM AMP-PNP was substituted in place of ATP. The samples were incubated for 10 minutes at room temperature, divided into four aliquots and transferred into 0.2 mL PCR tubes. Each sample was heated in parallel for 3 minutes to its respective temperature (51-64°C), as previous studies have indicated that a 3–5-minute exposure is optimal for visualizing protein denaturation (Martinez Molina et al., 2013 [DOI](#); Mateus et al., 2020 [DOI](#)). Subsequently, the samples were centrifuged at 180000g for 15 minutes at 4°C, and the supernatant was collected for in-solution digestion.

Detergent-based TPP on membrane proteomes

For detergent-based TPP (DB-TPP), ~2 mg of crude membranes from *Mus musculus* was solubilized in Solubilization Buffer for 30 min at 4 °C with gentle shaking. The insoluble material was pelleted by ultracentrifugation (180,000g, 15 min). The detergent-solubilized library was split into two equal aliquots representing the control and treatment samples. Both the treatment and control groups were supplemented with 5 mM MgCl₂ and 0.2 mM VO₄. Treatment samples were then exposed to a final concentration of 2 mM ATP, while the control received an equal volume of ddH₂O. The samples were incubated for 10 minutes at room temperature, divided into four aliquots and transferred into 0.2 mL PCR tubes. Each sample was heated in parallel for 3 minutes

to its respective temperature (51–64°C). Subsequently, the samples were centrifuged at 180,000g for 15 minutes at 4°C, and the supernatant was collected for processing as described by Johnston et al. (2022). Briefly, silica beads (9–13 µm diameter) were resuspended in water, washed once with 100% acetonitrile (CAN), rinsed twice with water, and resuspended at 50 mg/mL final. After each wash, the beads were isolated by brief centrifugation at 16,000 × g for 1 min. Crude membranes (~1 mg) were resuspended in ice-cold TS buffer (50 mM Tris-HCl, pH 8.0, 100 mM NaCl) supplemented with 1% (w/v) SDC for 30 min at 4°C with gentle shaking. The detergent extract was clarified by ultracentrifugation (110,000 × g for 15 min at 4°C), and aliquots (100 µg each) were gently vortex-mixed with glass beads (1 mg). CAN was then added to a final concentration of 80%, and samples were centrifuged at 16,000 × g for 5 min. The beads were rinsed three times with 500 µL of 80% ethanol without disturbing the pellet. After a final wash, the beads underwent MS sample preparation and LC-MS/MS analysis.

MS sample preparation and LC-MS/MS analysis

Equal volumes of supernatants from the treatment and control groups were treated with 6 M urea at room temperature for 30 min before reduction with 10 mM fresh dithiothreitol (DTT) for 1 h. Alkylation was performed with 20 mM iodoacetamide (IAA) in the dark at room temperature for 30 min, followed by a second round of reduction via 10 mM DTT for 30 min. The urea was diluted to 1 M with Buffer A. Trypsin digestion was performed with an enzyme/protein ratio of 1:100 at room temperature for 24 h. The tryptic peptides were acidified to pH 3 with 10% formic acid and desalted using hand-packed stage tips of C18 resin. The peptides were eluted with 80% acetonitrile and 0.1% formic acid and were dried by vacuum centrifugation. The analysis of tryptic peptides was performed in a NanoLC connected to an Orbitrap Exploris mass spectrometer (Thermo Fisher Scientific) was used for the analysis of all samples. The peptide separation was carried out using a Proxeon EASY nLC 1200 System (Thermo Fisher Scientific) fitted with a custom-made C18 column (15 cm × 150 µm ID) packed with HxSil C18 3 µm Resin 100 Å (Hamilton). A gradient of water/acetonitrile/0.1% formic acid was employed for chromatography. The samples were injected into the column and run for 180 minutes at a flow rate of 0.60 µL/min. The peptide separation began with 1% acetonitrile, increasing to 3% in the first 4 minutes, followed by a linear gradient from 3% to 23% acetonitrile over 86 minutes, then another increase from 24% to 80% acetonitrile over 35 minutes, and finally a 35-minute wash at 80% acetonitrile, and then decreasing to 1% acetonitrile for 10 min and kept 1% acetonitrile for another 10 min. The eluted peptides were ionized using positive nanoelectrospray ionization (NSI) and directly introduced into the mass spectrometer with an ion source temperature set at 250°C and an ion spray voltage of 2.1 kV. Full-scan MS spectra (m/z 350–2000) were captured in Orbitrap Exploris at a resolution of 120,000 (m/z 400). The automatic gain control was set to 1e6 for full FTMS scans and 5e4 for MS/MS scans. Ions with intensities above 1500 counts underwent fragmentation via nanoelectrospray ionization (NSI) in the linear ion trap. The top 15 most intense ions with charge states of ≥ 2 were sequentially isolated and fragmented using normalized collision energy of 30%, activation Q of 0.250, and an activation time of 10 ms. Ions selected for MS/MS were excluded from further selection for 3 seconds. The Orbitrap Exploris mass spectrometer was operated using Thermo Xcalibur software.

Data analysis in MaxQuant

Raw mass spectrometric files were analyzed in the MaxQuant environment, version 2.4.1.0. The MS/MS spectra were searched using the Andromeda search engine against the UniProt-mouse protein database (UP000000589, December 2021, 55086 entries) and UniProt-*Escherichia coli* protein database (UP000002032, July 2009, 4156 entries). Precursor mass and fragment mass were set with initial mass tolerances of 20 ppm for both the precursor and fragment ions. The search included variable modifications of asparagine/glutamine deamidation, methionine oxidation, and N-terminal acetylation and a fixed modification of carbamidomethyl cysteine. The maximum number of missed cleavages was set at two, and the maximum modifications/peptide and minimum peptide length were set at six amino acids. The UniProt database was also concatenated

with an automatically generated reverse database to estimate the false discovery rate (FDR) by using a target decoy search. The false discovery rate (FDR) was set at 0.01 for the peptide spectrum match (PSM) and protein identifications. When identified peptides were all shared between two proteins, they were combined and reported as one protein group. For relative quantification, MaxQuant's label-free quantification method, LFQ, was enabled.

Statistical analysis

Each treatment and control sample were collected from three technical replicates. The ProteinGroups.txt output file from MaxQuant was exported into Perseus v1.6.15.0 for downstream analysis. In-house functions of Perseus were used to identify and remove protein groups from the reverse decoy database, those marked as potential contaminants, or those only identified by a post-translational modification site. The remaining intensity, and label-free quantification (LFQ) intensity were $\log_2$ transformed and normalized to the mean peptide intensity value. To identify proteins with significant change in thermal stability, a Student's t-test was conducted with a within-groups variance, s_0 , set to 0.1. The test was applied to data filtered for proteins that had at least three valid LFQ intensities in either the treatment or control group. The remaining undefined intensity values were imputed from a normal distribution with a downshift of 1.8 standard deviations from the total sample mean and a width of 0.3 times the sample standard deviation. To determine significantly stabilized or destabilized as a result of ligand exposure, proteins with a $-\log_{10}$ p-value > 1.3 and LFQ peptide intensity $\log_2$ fold change > 1 or < -1 were considered as the direct or indirect candidates of ligand-binders. Only proteins with at least two unique peptides were considered for this calculation, a threshold selected because it aligns with established LC-MS/MS data analysis practices (The et al., 2020 [↗](#); Kurzawa et al., 2020; Reinhard et al., 2015 [↗](#)). A table of the unique peptide count for the discussed proteins in the manuscript is provided in Supplemental Table 2. To visualize the temperature-dependent peptide intensity data, the smoothed curves were generated using the Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) method. PCHIP interpolation was applied to the mean intensity values of each experimental group across the temperature gradient. Figures were generated through NumPy, Matplotlib, and Pandas Python coding language.

Ligand binding prediction with AlphaFold3

The web-based AlphaFold server (<https://alphafoldserver.com> [↗](#)) was utilized to predict ligand binding onto predicted protein structures. Protein peptide sequences were obtained from Uniprot for structure prediction. The resulting output data file was analyzed in Pymol to assess the corresponding B-factor/pLDDT score for each residue interacting with the ligand of interest. The mean pLDDT score from all interacting residues was determined to generate a confidence prediction of ligand binding.

Protein annotation

The protein list obtained from MaxQuant was subjected to a Gene Ontology (GO) term analysis using the UniProtKB database to identify proteins with the GO-term “membrane”. Proteins from this group were extracted in FastA format and the Phobius web server (<http://phobius.sbc.su.se/> [↗](#)) was utilized to predict the number of transmembrane segments (TMS). Any protein with at least one TMS was classified as an IMP. Protein with no TMS but with the GO annotation “membrane” was classified as a membrane-associated protein; all other proteins were considered as soluble proteins. To assess molecular functions of significantly stabilized IMPs, the Gene Ontology Molecular Function (“GO_MF_Direct/GO_MF_FAT”) was used through DAVID Bioinformatics (<https://david.ncifcrf.gov/> [↗](#)). An EASE score of 0.05 was applied to test for significant GO terms based on a p-value cut-off of 0.05 after Benjamini-Hochberg correction. The gProfiler g:GOST tool was utilized to reduce redundancy of significantly enriched terms (<https://biit.cs.ut.ee/gprofiler/gost> [↗](#)).

Data availability statement

The MS-based proteomics data of this study have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository and are available through the identifier PXD055093.

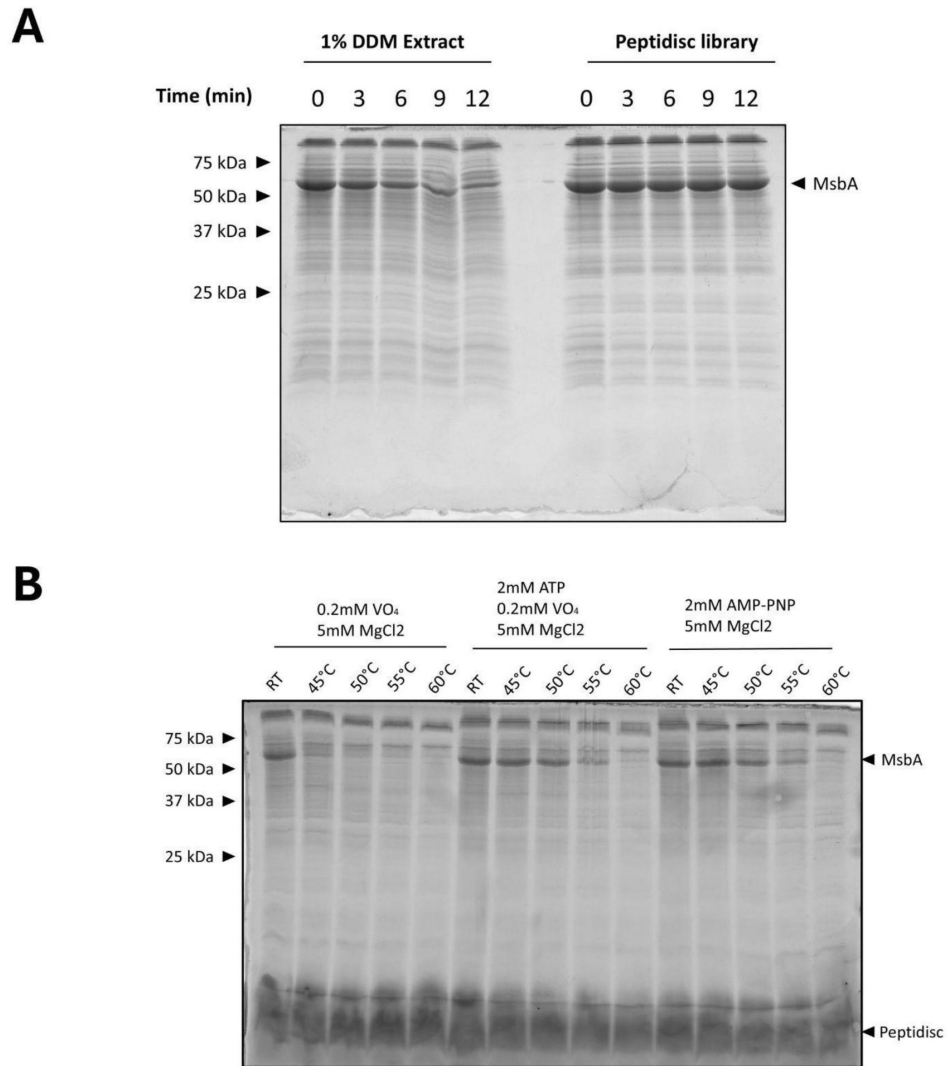
Acknowledgements

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Additional files

Supplementary File 1. [🔗](#)

Supplementary figures and tables



Supplementary Figure 1

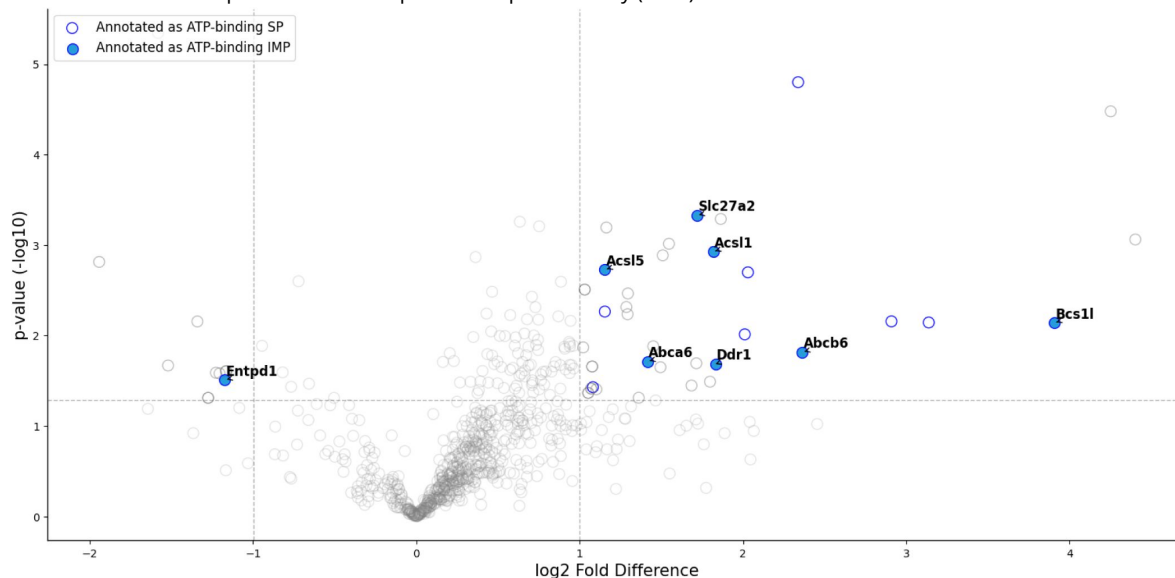
Thermal stability of MsbA in detergent and in Peptidisc with different ligands.

(A) The *E. coli* membrane fraction enriched for MsbA was solubilized with 1% DDM or reconstituted in Peptidisc. The detergent extract and Peptidisc library were incubated at 45°C for the indicated times. After ultra-centrifugation, the supernatants were analyzed on 15% SDS-PAGE and visualized with Coomassie blue staining. (B) The Peptidisc library prepared in (A) was incubated with ATP- VO_4 or AMP-PNP at the indicated temperature. After ultra-centrifugation, the supernatants were analyzed on 15% SDS-PAGE and visualized with Coomassie blue staining.

Supplementary Figure 2

Volcano plot of stabilized and destabilized proteins at 51 °C in the presence of AMP-PNP.

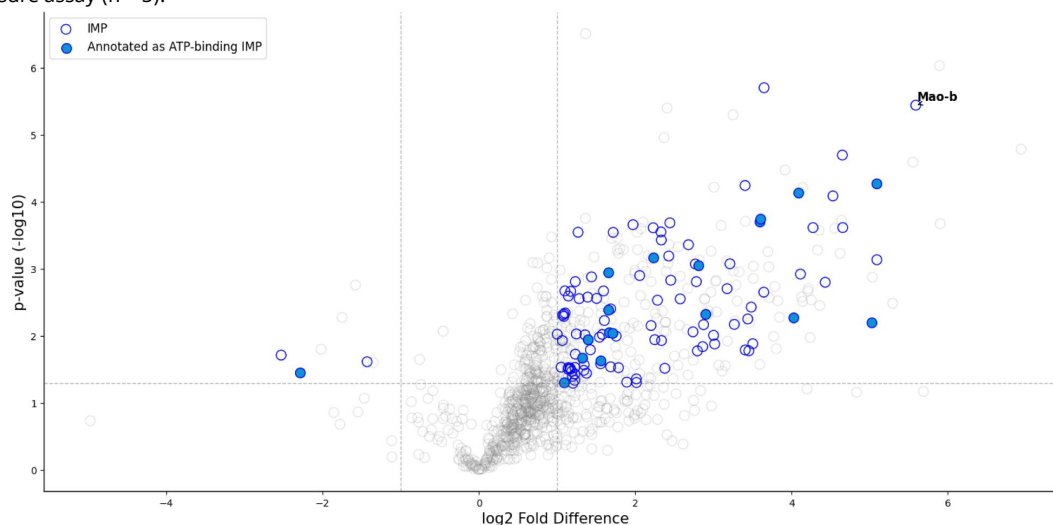
Soluble proteins (SP) annotated as ATP-binding proteins are presented as hollow blue circles, and IMPs annotated as ATP-binding are presented as solid blue circles. Values taken from $\log_2$ -transformed LFQ peptide intensities. The mean value is obtained from three replicates at the temperature exposure assay ($n = 3$).

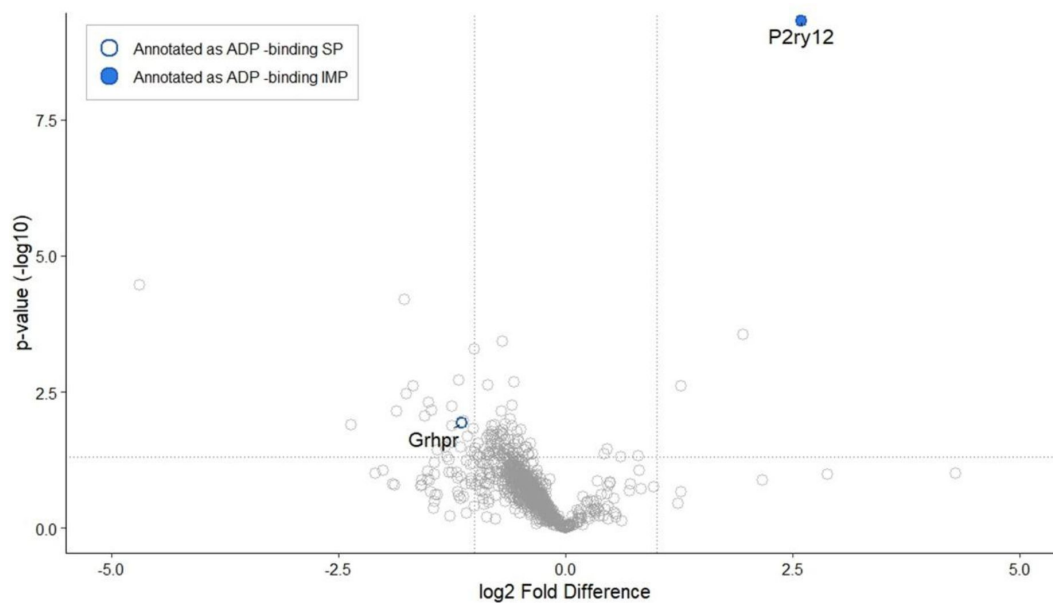


Supplementary Figure 3

Volcano plot of stabilized and destabilized proteins at 64 °C in the presence of ATP-VO₄.

IMPs are represented by hollow blue circles and IMPs annotated as ATP-binding are represented by solid blue circles. Values taken from $\log_2$ -transformed LFQ peptide intensities. The mean value is obtained from three replicates at the temperature exposure assay ($n = 3$).

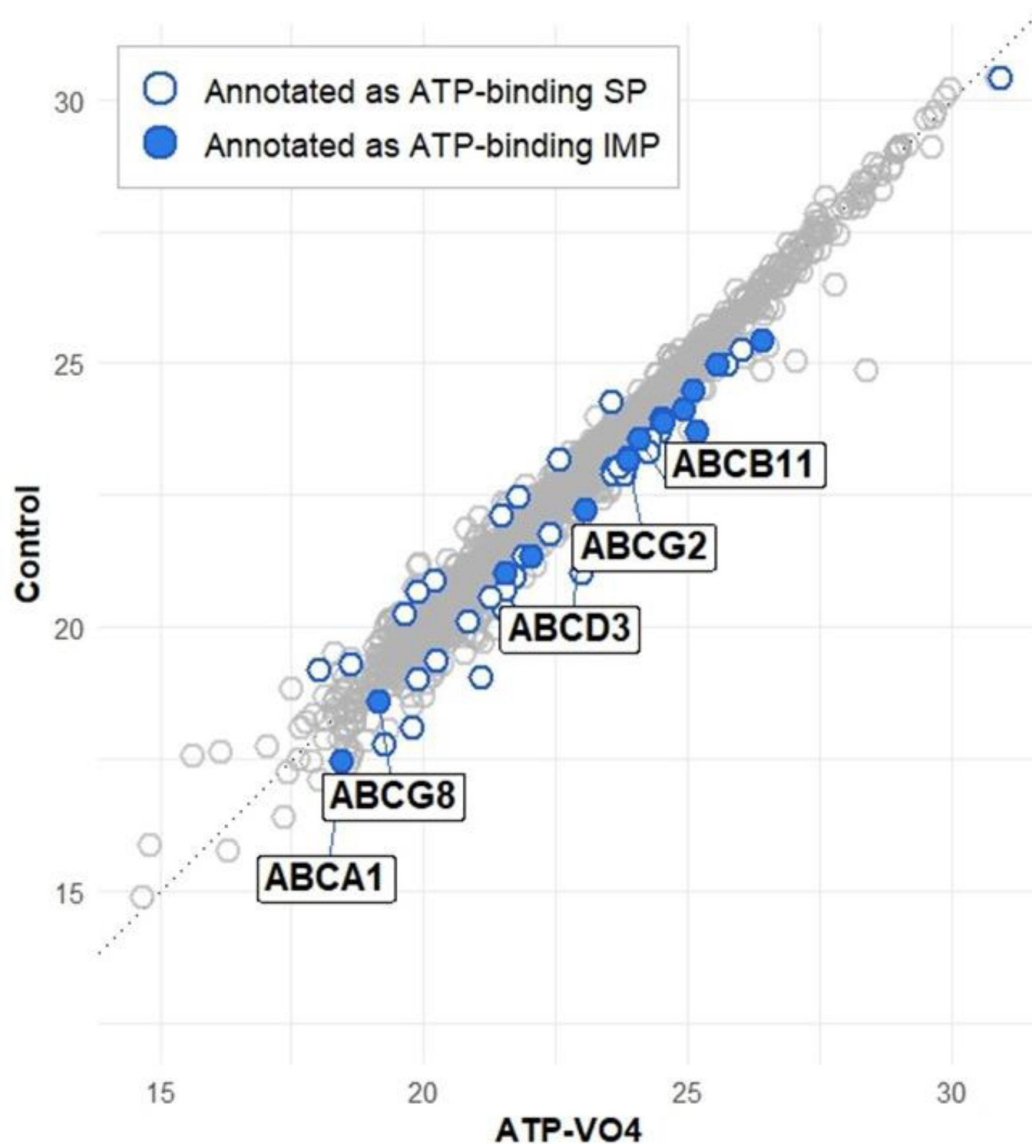




Supplementary Figure 4

Volcano plot of stabilized and destabilized proteins at 57 °C in the presence of 2-MeS-ADP.

The soluble proteins (SP) annotated as ADP-binding are represented by hollow blue circles and the IMPs annotated as ADP-binding are represented by solid blue circles. Values taken from $\log_2$ -transformed LFQ peptide intensities. The mean value is obtained from three replicates at the temperature exposure assay ($n = 3$).



Supplementary Figure 5

Correlation plot of protein differential stabilization at 51 °C in MM-TPP, performed with DIA data acquisition from a whole mouse liver library.

ATP-binding proteins with $|\log_2 \text{fold change}| \geq 0.5$ are highlighted. Soluble proteins (SP) annotated as ATP-binding are shown as hollow blue circles, while IMPs annotated as ATP-binding are shown as solid blue circles. ABC transporters that pass the significance threshold are labeled on the plot.

Temperature	Total protein	Total IMP
Room temperature	998 ± 2	470 ± 3 (47%)
45°C	892 ± 11	425 ± 1 (48%)
50°C	810 ± 16	385 ± 7 (48%)
55°C	734 ± 46	354 ± 22 (48%)
60°C	566 ± 33	276 ± 11 (49%)
65°C	524 ± 4	264 ± 3 (50%)

Supplemental Table 1

Total and Integral Membrane Protein Counts Across Temperatures in Mouse Liver Peptidisc Libraries.

The % value is used to assess IMP loss relative to total protein count. Each protein was identified based on at least two unique peptides (n = 2)

Target	Treatment	Number of unique peptides			
		51 °C	56/57 °C	60/61 °C	64 °C
MsbA	ATP-VO ₄	Control: 13,12,6	Control:11,4,9	Control:7,5,6	

Supplemental Table 2

Unique peptide counts for proteins highlighted in this study.

MsbA, ABCA6, ABCB1, ABCB6, ABCC2, ABCC3, ABCC9, ABCG2, ABCG5, BCS1L, P2RX4, P2RY6, P2RY12, and MAO-B. Counts are from MaxQuant analyses using a data-dependent acquisition (DDA) workflow across the temperatures tested in MM-TPP (with ATP- VO₄, 2-MeS-ADP, and AMP-PNP) and in DB-TPP (with ATP-VO₄).

		Treatment: 16,16,15	Treatment:17,15,17	Treatment: 14,11,14	
ABCA6	ATP-VO ₄	Control: 9,12,13	Control:5,3,6	Control:1,0,3	Control:1,0,0
		Treatment:21,19,23	Treatment:12,15,17	Treatment:12,13,11	Treatment:7,7,9
	ATP-VO ₄ (DB-TPP)	Control:22,25,25	Control:25,27,23	Control:22,23,22	Control:14,19,15
		Treatment:25,25,24	Treatment:22,24,23	Treatment:17,5,22	Treatment:13,20,15
	2-MeS- ADP	Control:6,5,5	Control:0,1,0		
		Treatment:1,5,7	Treatment:2,2,1		
	AMP-PNP	Control:6,5,5	Control:3,1,5	Control:1,2,3	Control:0,0,0
		Treatment:6,7,6	Treatment:3,6,4	Treatment:2,0,1	Treatment:0,0,0
	ATP-VO ₄	Control:13,10,15	Control:0,3,4	Control:1,0,0	Control:0,1,0
		Treatment:21,22,21	Treatment:6,13,12	Treatment:6,10,9	Treatment:1,2,3
ABCB11	ATP-VO ₄ (DB-TPP)	Control:30,32,29	Control:26,25,27	Control:22,20,24	Control:16,15,16
		Treatment:38,37,38	Treatment:30,32,23	Treatment:20,5,19	Treatment:15,23,17
	2-MeS- ADP	Control:0,0,0	Control:0,0,0		
		Treatment:0,0,0	Treatment:0,0,0		
	AMP-PNP	Control:6,6,5	Control:1,2,3	Control:1,2,1	Control:0,0,0

Supplemental Table 2 (continued)

		Treatment:8,8,7	Treatment:7,3,4	Treatment:2,0,4	Treatment:0,0,0
ABCB6	ATP-VO ₄	Control:0,1,1	Control:0,0,0	Control:0,0,0	Control:0,0,0
		Treatment:3,3,3	Treatment:0,0,0	Treatment:0,0,0	Treatment:0,0,0
	ATP-VO ₄ (DB-TPP)	Control:0,0,0	Control:0,0,0	Control:0,0,0	Control:0,0,0
		Treatment:0,0,0	Treatment:0,0,0	Treatment:	Treatment:0,0,0
	2-MeS- ADP	Control:0,0,0	Control:0,0,0		
		Treatment:0,0,0	Treatment:0,0,0		
	AMP-PNP	Control:0,0,0	Control:0,0,0	Control:0,0,0	Control:0,0,0
		Treatment:3,2,2	Treatment:2,2,2	Treatment:0,0,0	Treatment:0,0,0
ABCC2	ATP-VO ₄	Control:6,4,9	Control:0,0,0	Control:0,0,0	Control:0,0,0
		Treatment:13,15,10	Treatment:6,6,5	Treatment:3,5,2	Treatment:0,0,0
	ATP-VO ₄ (DB-TPP)	Control:13,15,15	Control:16,11,12	Control:8,10,8	Control:3,3,5
		Treatment:22,19,16	Treatment:17,16,15	Treatment:7,6,11	Treatment:8,12,6
	2-MeS- ADP	Control:2,2,4	Control:0,0,0		
		Treatment:3,1,6	Treatment:1,2,1		
	AMP-PNP	Control:1,1,3	Control:0,0,0	Control:0,0,0	Control:0,0,0
		Treatment:4,3,2	Treatment:0,0,0	Treatment:0,0,0	Treatment:0,0,0

Supplemental Table 2 (continued)

ABCC3	ATP-VO ₄	Control:10,6,7	Control:0,0,0	Control:0,0,0	Control:0,0,0
		Treatment:13,15,15	Treatment:0,0,0	Treatment:2,3,5	Treatment:0,0,0
	ATP-VO ₄ (DB-TPP)	Control:0,0,0	Control:0,0,0	Control:0,0,0	Control:0,0,0
		Treatment:0,0,0	Treatment:0,0,0	Treatment:0,0,0	Treatment:0,0,0
	2-MeS-ADP	Control:0,0,0	Control:0,0,0		
		Treatment:0,0,0	Treatment:0,0,0		
	AMP-PNP	Control:0,0,0	Control:0,0,0	Control:0,0,0	Control:0,0,0
		Treatment:0,0,0	Treatment:0,0,0	Treatment:0,0,0	Treatment:0,0,0
	ATP-VO ₄	Control:15,14,13	Control:6,5,6	Control:2,0,4	Control:1,0,2
		Treatment:15,16,17	Treatment:5,8,9	Treatment:7,13,10	Treatment:8,7,6
ABCC9	ATP-VO ₄	Control:1,1,1	Control:1,1,1	Control:1,1,1	Control:1,1,1
		Treatment:1,1,1	Treatment:1,1,1	Treatment:1,1,1	Treatment:1,1,1
	ATP-VO ₄ (DB-TPP)	Control:1,1,1	Control:1,1,1	Control:1,1,1	Control:1,1,1
		Treatment:1,1,1	Treatment:1,1,1	Treatment:1,1,1	Treatment:1,1,1
	2-MeS-ADP	Control:16,19,18	Control:6,8,3		
		Treatment:15,18,20	Treatment:3,12,8		
	AMP-PNP	Control:13,12,11	Control:5,7,7	Control:5,9,6	Control:4,3,6
		Treatment:13,13,13	Treatment:11,9,9	Treatment:7,3,8	Treatment:3,5,4
	ATP-VO ₄	Control:15,14,13	Control:6,5,6	Control:2,0,4	Control:1,0,2
		Treatment:15,16,17	Treatment:5,8,9	Treatment:7,13,10	Treatment:8,7,6

Supplemental Table 2 (continued)

ABCG2	ATP-VO ₄	Control:6,8,9	Control:6,6,5	Control:2,0,1	Control:0,0,1
		Treatment:8,7,9	Treatment:8,8,7	Treatment:9,8,7	Treatment:4,5,5
	ATP-VO ₄ (DB-TPP)	Control:10,11,11	Control:9,9,10	Control:5,7,6	Control:4,4,4
		Treatment:10,11,1 2	Treatment:10,10,7	Treatment:3,1,5	Treatment:2,6,4
	2-MeS- ADP	Control:4,3,3	Control:1,1,0		
		Treatment:3,3,3	Treatment:0,1,0		
	AMP-PNP	Control:3,3,3	Control:1,0,0	Control:0,0,0	Control:0,0,0
		Treatment:3,3,3	Treatment:3,3,3	Treatment:3,2,3	Treatment:0,0,0
	ATP-VO ₄	Control:1,1,1	Control:0,0,0	Control:0,0,0	Control:0,0,0
		Treatment:2,2,2	Treatment:0,0,0	Treatment:0,0,0	Treatment:0,0,0
ABCG5	ATP-VO ₄	Control:2,1,2	Control:1,1,1	Control:0,0,0	Control:0,0,0
		Treatment:2,2,1	Treatment:1,0,1	Treatment:0,0,0	Treatment:0,0,0
	2-MeS- ADP	Control:0,0,0	Control:0,0,0		
		Treatment:0,0,0	Treatment:0,0,0		
	AMP-PNP	Control:0,0,0	Control:0,0,0	Control:0,0,0	Control:0,0,0
		Treatment:0,0,0	Treatment:0,0,0	Treatment:0,0,0	Treatment:0,0,0

Supplemental Table 2 (continued)

BCS11	ATP-VO ₄	Control: 0,0,0	Control:0,0,0	Control: 0,0,0	Control: 0,0,0
		Treatment: 6,7,6	Treatment: 1,4,2	Treatment:0,0,0	Treatment: 1,0,0
	ATP-VO ₄ (DB-TPP)	Control: 1,4,4	Control:1,0,1	Control: 0,0,0	Control: 0,0,0
		Treatment: 7,5,4	Treatment: 0,2,3	Treatment: 0,0,0	Treatment: 1,0,0
	2-MeS- ADP	Control: 0,0,0	Control: 0,0,0		
		Treatment:,0,0,0	Treatment: 0,0,0		
	AMP-PNP	Control: 0,0,1	Control:0,0,0	Control:0,0,0	Control:0,0,0
		Treatment:13.10.1 2	Treatment:15,15,15	Treatment:15,4,12	Treatment:16,13,9
	ATP-VO ₄	Control: 3, 2, 3	Control: 3,3,2	Control: 0,0,0	Control:3,4,5
		Treatment: 3,3,1	Treatment:1,0,1	Treatment: 0,0,0	Treatment:0,1,3
P2RX4	ATP-VO ₄ (DB-TPP)	Control: 3,1,4	Control: 2,3,2	Control: 4,3,3	Control:1,1,2
		Treatment:2,1,1	Treatment: 1,1,2	Treatment:1,2,2	Treatment: 1,2,2
	2-MeS- ADP	Control: 0,0,0	Control: 0,0,0		
		Treatment: 0,0,0	Treatment: 0,0,0		
	AMP-PNP	Control:2,2,2	Control:2,2,2	Control:2,2,2	Control:2,2,2
		Treatment:2,1,2	Treatment:2,1,1	Treatment:2,0,2	Treatment:2,2,2

Supplemental Table 2 (continued)

P2RY6	ATP-VO ₄	Control: 12, 13, 13	Control:2,2,3	Control:1,0,0	Control:0,0,0
		Treatment:12,10,13	Treatment:3,5,6	Treatment:2,4,3	Treatment:0,0,0
	ATP-VO ₄ (DB-TPP)	Control: 22,22,22	Control: 4,2,3	Control:0,0,1	Control: 0,1,0
		Treatment: 25,23,25	Treatment: 3,1,5	Treatment: 0,0,1	Treatment: 1,0,0
	2-MeS-ADP	Control: 2,2,2	Control: 1,1,0		
		Treatment: 0,2,2	Treatment: 1,1,1		
	AMP-PNP	Control:7,5,6	Control:1,0,0	Control:0,0,0	Control: 0,0,0
		Treatment:8,8,4	Treatment:1,2,3	Treatment: 0,0,0	Treatment: 0,0,0
	ATP-VO ₄	Control:19,15,15	Control:2,5,4	Control:0,0,1	Control:0,0,0
		Treatment:14,14,13	Treatment:3,6,9	Treatment: 8,9,8	Treatment: 5,4,5
P2RY12	ATP-VO ₄ (DB-TPP)	Control:29, 30, 30	Control: 22, 21, 25	Control: 20, 16, 18	Control:14, 12, 14
		Treatment:29, 28, 29	Treatment: 23, 21, 23	Treatment: 19,11,17	Treatment:13,16,14
	2-MeS-ADP	Control: 6,4,7	Control: 2,3,4		
		Treatment: 7,6,7	Treatment: 4,3,6		
	AMP-PNP	Control:2,1,2	Control:0,0,0	Control:0,0,0	Control:0,0,0
		Treatment:3,2,2	Treatment:0,0,0	Treatment:0,0,0	Treatment:0,0,0

Supplemental Table 2 (continued)

Mao-B	ATP-VO ₄	Control:20,18,18	Control:15,14,14	Control:14,0,13	Control:1,1,1
		Treatment:20,21,20	Treatment:16,14,17	Treatment:14,17,18	Treatment:14,15,15
	ATP-VO ₄ (DB-TPP)	Control:25,24,26	Control:25,24,27	Control:21,19,21	Control:15,15,18
		Treatment:26,23,26	Treatment:25,21,21	Treatment:17,6,19	Treatment:14,17,18
	2-MeS-ADP	Control:23,21,22	Control:20,17,17		
		Treatment:18,24,24	Treatment:20,18,19		
	AMP-PNP	Control:15,17,16	Control:10,13,13	Control:13,13,14	Control:8,9,8
		Treatment:17,17,17	Treatment:15,12,13	Treatment:10,10,14	Treatment:10,9,7

Supplemental Table 2 (continued)

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Reviewer #1 (Public review):

Summary:

The idea is appealing, but the authors have not sufficiently demonstrated the utility of this approach.

Strengths:

Novelty of the approach, potential implications for discovering novel interactions

Comments on revisions:

The authors have adequately addressed most of my concerns in this improved version of the manuscript

<https://doi.org/10.7554/eLife.104549.2.sa2>

Reviewer #2 (Public review):

Summary:

The membrane mimetic thermal proteome profiling (MM-TPP) presented by Jandu et al. promises a useful way to minimize the interference of detergents in efficient mass spectrometry analysis of membrane proteins. Thermal proteome profiling is a mass spectrometric method that measures binding of a drug to different proteins in a cell lysate by monitoring thermal stabilization of the proteins because of the interaction with the ligands that are being studied. This method has been underexplored for membrane proteome because of the inefficient mass spectrometric detection of membrane proteins and because of the interference from detergents that are used often for membrane protein solubilization.

Strengths:

In this report the binding of ligands to membrane protein targets has been monitored in crude membrane lysates or tissue homogenates exalting the efficacy of the method to detect both intended and off-target binding events in a complex physiologically relevant sample setting. The manuscript is lucidly written and the data presented seems clear. Kudos to the authors. This methodology shows immense potential for identifying membrane protein binders (small-molecule or protein) in a near-native environment, and as a result promises to be a great tool for drug discovery campaigns.

Weaknesses:

While this is a solid report and a promising tool for analyzing membrane protein drug interactions in a detergent-free environment, it is crucial to bear in mind that the process of reconstitution begins with detergent solubilization of the proteome and does not completely circumvent structural perturbations invoked by detergents.

<https://doi.org/10.7554/eLife.104549.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Summary:

The idea is appealing, but the authors have not sufficiently demonstrated the utility of this approach.

Strengths:

Novelty of the approach, potential implications for discovering novel interactions

Weaknesses:

The Duong had introduced their highly elegant peptidisc approach several years ago. In this present work, they combine it with thermal proteome profiling (TPP) and attempt to demonstrate the utility of this combination for identifying novel membrane protein-ligand interactions.

While I find this idea intriguing, and the approach potentially useful, I do not feel that the authors had sufficiently demonstrated the utility of this approach. My main concern is that no novel interactions are identified and validated. For the presentation of any new methodology, I think this is quite necessary. In addition, except for MsbA, no orthogonal methods are used to support the conclusions, and the authors rely entirely on quantifying rather small differences in abundances using either iBAQ or LFQ.

We thank the reviewer for their thoughtful comments. In this revision, we have experimentally addressed the reviewer's concerns in three ways:

(1) To demonstrate the utility of our MM-TPP method over the detergent-based TPP workflow (termed DB-TPP), we performed a side-by-side comparison using ATP-VO₄ at 51 °C (Figure 3B and Figure 4A). From the DB-TPP dataset, 7.4% of all identified proteins were annotated as ATP-binding, while 6.4% of proteins differentially stabilized were annotated as ATP-binding. In contrast, in the MM-TPP dataset, 9.3% of all identified proteins were annotated as ATP-binding proteins, while 17% of proteins differentially stabilized were annotated as ATP-binding. The lack of enrichment in the detergent-based approach indicates that the observed differences are likely stochastic, rather than a result of specific ATP-VO₄-mediated stabilization as found with MM-TPP. For instance, several key proteins—BCS1, P2RY6, SLC27A2, ABCB1, ABCC2, and ABCC9—found differentially stabilized using the MM-TPP method showed no such pattern in the DB-TPP dataset. This divergence strongly supports the specificity and utility of our Peptidisc approach.

(2) To demonstrate that MM-TPP can resolve not only the broader effects of ATP-VO₄ but also specific ligand-protein interactions, we employed 2-methylthio-ADP (2-MeS-ADP), a selective agonist of the P2RY12 receptor [PMID: 24784220]. In that case, we observed clear thermal stabilization of P2RY12, with more than 6-fold increase in stability at both 51 °C and 57 °C (–log₁₀ p > 5.97; Figure 4B and Figure S4). Notably, no other proteins—including the structurally related but non-responsive P2RY6 receptor—showed comparable stabilization fold change at these temperatures.

(3) To further probe the reproducibility of the method, we performed an independent MMTTP evaluation with ATP-VO₄ at 51 °C using data-independent acquisition (DIA), in contrast to the data-dependent acquisition (DDA) approach used in the initial study (Figure S5). Overall, 7.8% of all identified proteins were annotated as ATP-binding, and as before, this proportion

increased to 17% among proteins with $\log_2$ fold changes greater than 0.5. Specifically, BCS1 and SLC27A2 exhibited strong stabilization ($\log_2$ fold change > 1), while P2RY6, ABCB11, ABCC2, and ABCG2 showed moderate stabilization ($\log_2$ fold changes between 0.5 and 1), and consistent with previous results, P2RX4 was destabilized, with a $\log_2$ fold change below -1. These findings support the consistency and reproducibility of the method across distinct data acquisition methods.

My main concern is that no novel interactions are identified and validated. For the presentation of any new methodology, I think this is quite necessary.

The primary objective of our study is to establish and benchmark the MM-TPP workflow using known targets, rather than to discover novel ligand–protein interactions. Identifying new binders requires extensive screening and downstream validations, which we believe is beyond the scope of this methodological report. Instead, our study highlights the sensitivity and reliability of the MM-TPP approach by demonstrating consistent and reproducible results with well-characterized interactions.

We respectfully disagree with the notion that introducing a new methodology must necessarily include the discovery of novel interactions. For instance, Martinez Molina et al. [PMID: 23828940] introduced the cellular thermal shift assay (CETSA) by validating established targets such as MetAP2 with TNP-470 and CDK2 with AZD-5438, without identifying novel protein–ligand pairs. Similarly, Kalxdorf et al. [PMID: 33398190] published their cell-surface thermal proteome profiling (CS-TPP) using Ouabain to stabilize the Na^+/K^+ -ATPase pump in K562 cells, and SB431542 to stabilize its canonical target JAG1. In fact, when these methods revealed additional stabilizations, these were not validated but instead interpreted through reasoning grounded in the literature. For instance, they attributed the SB431542-induced stabilization of MCT1 to its reported role in cell migration and tumor invasiveness, and explained that SLC1A2 stabilization is related to the disruption of Na^+/K^+ -ATPase activity by Ouabain. In the same way, our interpretation of ATP- VO_4 -mediated stabilization of Mao-B is justified by predictive AlphaFold-3 rather than direct orthogonal assays, which are beyond the scope of our methodological presentation.

Collectively, the influential studies cited above have set methodological precedents by prioritizing validation and proof-of-concept over merely finding uncharacterized binders. In the same spirit, our work is centred on establishing MM-TPP as a robust platform for probing membrane protein–ligand interactions in a water-soluble format. The discovery of novel binders remains an exciting future direction—one that will build upon the methodological foundation laid by the present study.

In addition, except for MsbA, no orthogonal methods are used to support the conclusions, and the authors rely entirely on quantifying rather small differences in abundances using either iBAQ or LFQ.

We deliberately began this study with our model protein, MsbA, examined under both native and overexpressed conditions, to establish an adequation between MMTTP (Figure 2D) and biochemical stability assays (Figure 2A). This validation has provided us with the foundation to confidently extend MM-TPP to the mouse organ proteome. To demonstrate the validity of our workflow, we have used ATP- VO_4 because it has expected targets.

We note that orthogonal validation often requires overproduction and purification of the candidate proteins, including suitable antibodies, which is a true challenge for membrane proteins. Here, we demonstrate that MM-TPP can detect ligand-induced thermal shifts directly in native membrane preparations, without requiring protein overproduction or purification. We also emphasize several influential studies in TPP, including Martinez Molina et al. (PMID: 23828940) and Fang et al. (PMID: 34188175), which focused primarily on

establishing and benchmarking the methodology, rather than on extensive orthogonal validation. In the same spirit, our study prioritizes methodological development, and accordingly, several orthogonal validations are now included in this revision.

[...] and the authors rely entirely on quantifying rather small differences in abundances using either iBAQ or LFQ.

To clarify, all analyses on ligand-induced stabilization or destabilization were carried out using LFQ values. The sole exception is on Figure 2B, where we used iBAQ values to depict the relative abundance of proteins within a single sample; this to show MsbA's relative level within the *E. coli* peptidisc library.

Respectfully, we disagree with the assertion that we are “quantifying rather small differences in abundances using either iBAQ or LFQ.” We were able to clearly distinguish between stabilizations driven by specific ligands binding to their targets versus those caused by non-specific ligands with broader activity. This is further confirmed by comparing 2-MeS-ADP, a selective ligand for P2RY12, with ATP-VO₄, a highly promiscuous ligand, and AMP-PNP, which exhibits intermediate breadth. When tested in triplicate at 51 °C, 2-MeS-ADP significantly altered the thermal stability of 27 proteins, AMP-PNP 44 proteins, and ATP-VO₄ 230 proteins, consistent with the expectation that broader ligands stabilize more proteins nonspecifically. Importantly, 2-MeS-ADP produced markedly stronger stabilization of its intended target, P2RY12 ($-\log_{10}p = 9.32$), than the top stabilized proteins for ATP-VO₄ (DNAJB3, $-\log_{10}p = 5.87$) or AMP-PNP (FTH1, $p = 5.34$). Moreover, 2-MeS-ADP did not significantly stabilize proteins that were consistently stabilized by the broad ligands, such as SLC27A2, which was strongly stabilized by both ATP-VO₄ and AMP-PNP ($-\log_{10}p > 2.5$). Together, these findings demonstrate that MMTTP can robustly distinguish between broad-spectrum and target-specific ligands, with selective ligands inducing stronger and more physiologically meaningful stabilization at their intended targets compared to promiscuous ligands.

Finally, we emphasize that our findings are not marginal, but meet quantitative and statistical rigor consistent with best practices in proteomics. We apply dual thresholds combining effect size ($|\log_2FC| \geq 1$, i.e., at least a two-fold change) with statistical significance (FDR-adjusted $p \leq 0.05$)—criteria commonly used in proteomics methodology studies (e.g., PMID: 24942700, 38724498). Moreover, the stabilization and destabilization events we report are reproducible across biological replicates ($n = 3$), consistent across adjacent temperatures for most targets, and technically robust across acquisition modes (DDA vs. DIA). Taken together, these results reflect statistically valid and biologically meaningful effects, fully aligned with standards set by prior published proteomics studies.

Furthermore, the reported changes in abundances are solely based on iBAQ or LFQ analysis. This must be supported by a more quantitative approach such as SILAC or labeled peptides. In summary, I think this story requires a stronger and broader demonstration of the ability of peptidisc-TPP to identify novel physiologically/pharmacologically relevant interactions.

With respect to labeling strategies, we deliberately avoided using TMT due to concerns about both cost and potential data quality issues. Some recent studies have documented the drawbacks of TMT in contexts directly relevant to our work. For example, a benchmarking study of LIP-MS workflows showed that although TMT increased proteome depth and reduced technical variance, it was less accurate in identifying true drug–protein interactions and produced weaker dose–response correlations compared with label-free DIA approaches [PMID: 40089063]. More broadly, technical reviews have highlighted that isobaric tagging is intrinsically prone to ratio compression and reporterion interference due to co-isolation and co-fragmentation of peptides, which flatten measured fold-changes and obscure biologically meaningful differences [PMID: 22580419, 22036744]. In terms of SILAC, the technique

requires metabolic incorporation of heavy amino acids, which is feasible in cultured cells but not in physiologically relevant tissues such as the liver organ used here. SILAC mouse models exist, but they are expensive and time-consuming [PMID: 18662549, 21909926]. We are not a mouse lab, and introducing liver organ SILAC labeling in our workflow is beyond the scope of these revisions. We also note that several hallmark TPP studies have been successfully carried out using label-free quantification [PMID: 25278616, 26379230, 33398190, 23828940], establishing this as an accepted and widely applied approach in the field.

To further support our conclusions, we added controls showing that detergent solubilization of mouse liver membranes followed by SP4 cleanup fails to detect ATP-VO₄⁻ mediated stabilization of ATP-binding proteins, underscoring the necessity of Peptidisc reconstitution for capturing ligand-induced thermal stabilization. We also present new data demonstrating selective stabilization of the P2Y₁₂ receptor by its agonist 2-MeS-ADP, providing orthogonal, receptor-specific validation within the MM-TPP framework. Finally, an orthogonal DIA acquisition on separate replicates confirmed robust ATP-vanadate stabilization of ATP-binding proteins, including BCS1l and SLC27A2. Together, these additions reinforce that the observed stabilizations are genuine, physiologically relevant ligand–protein interactions and highlight the unique advantage of the Peptidisc-based workflow in capturing such events.

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Reviewer #2 (Public Review):

Summary:

The membrane mimetic thermal proteome profiling (MM-TPP) presented by Jandu et al. seems to be a useful way to minimize the interference of detergents in efficient mass spectrometry analysis of membrane proteins. Thermal proteome profiling is a mass spectrometric method that measures binding of a drug to different proteins in a cell lysate by monitoring thermal stabilization of the proteins because of the interaction with the ligands that are being studied. This method has been underexplored for membrane proteome because of the inefficient mass spectrometric detection of membrane proteins and because of the interference from detergents that are used often for membrane protein solubilization.

Strengths:

In this report the binding of ligands to membrane protein targets has been monitored in crude membrane lysates or tissue homogenates exalting the efficacy of the method to detect both intended and off-target binding events in a complex physiologically relevant sample setting.

The manuscript is lucidly written and the data presented seems clear. The only insignificant grammatical error I found was that the 'P' in the word peptidisc is not capitalized in the beginning of the methods section "MM-TPP profiling on membrane proteomes". The clear writing made it easy to understand and evaluate what has been presented. Kudos to the authors.

Weaknesses:

While this is a solid report and a promising tool for analyzing membrane protein drug interactions, addressing some of the minor caveats listed below could make it much more impactful.

The authors claim that MM-TPP is done by "completely circumventing structural perturbations invoked by detergents[1] ". This may not be entirely accurate, because before reconstitution of the membrane proteins in peptidisc, the membrane fractions are solubilized by 1% DDM. The solubilization and following centrifugation step lasts at least for 45 min. It is less likely that all the structural perturbations caused by DDM to various membrane proteins and their transient interactions become completely reversed or rescued by peptidisc reconstitution.

We thank the reviewer for this insightful comment. In response, we have revised the sentence and expanded the discussion to clarify that the Peptidisc provides a complementary approach to detergent-based preparations for studying membrane proteins, preserving native lipid–protein interactions and stabilization effects that may be diminished in detergent.

To further address the structural perturbations invoked by detergents, and as already detailed to our response to Reviewer 1, we have compared the thermal profile of the Peptidisc library to the mouse liver membranes solubilized with 1% DDM, after incubation with ATP-VO₄ at 51 °C (Figure 4A). The results with the detergent extract revealed random patterns of stabilization and destabilization, with only 6.4% of differentially stabilized proteins being ATP-binding—comparable to the 7.4% observed in the background. In contrast, in the Peptidisc library, 17% of differentially stabilized proteins were ATP-binding, compared to 9.3% in the background. Thus, while Peptidisc reconstitution does not fully avoid initial detergent exposure, these findings underscore the importance of implementing Peptidisc in the TPP workflow when dealing with membrane proteins.

In the introduction, the authors make statements such as "...it is widely acknowledged that even mild detergents can disrupt protein structures and activities, leading to challenges in accurately identifying drug targets.." and "[peptidisc] libraries are instrumental in capturing and stabilizing IMPs in their functional states while preserving their interactomes and lipid allosteric modulators...". These need to be rephrased, as it has been shown by countless studies that even with membrane protein suspended in micelles robust ligand binding assays and binding kinetics have been performed leading to physiologically relevant conclusions and identification of protein-protein and protein-ligand interactions.

We thank the reviewer for this valuable feedback and fully agree with the point raised. In response, we have revised the Introduction and conclusion to moderate the language concerning the limitations of detergent use. We now explicitly acknowledge that numerous studies have successfully used detergent micelles for ligand-binding assays and kinetic analyses, yielding physiologically relevant insights into both protein–protein and protein–ligand interactions [e.g., PMID: 22004748, 26440106, 31776188].

At the same time, we clarify that the Peptidisc method offers a complementary advantage, particularly in the context of thermal proteome profiling (TPP), which involves mass spectrometry workflows that are incompatible with detergents. In this setting, Peptidiscs facilitate the detection of ligand-binding events that may be more difficult to observe in detergent micelles.

We have reframed our discussion accordingly to present Peptidiscs not as a replacement for detergent-based methods, but rather as a complementary tool that broadens the available methodological landscape for studying membrane protein interactions.

If the method involves detergent solubilization, for example using 1% DDM, it is a bit disingenuous to argue that 'interactomes and lipid allosteric modulators' characterized

by lowaffinity interactions will remain intact or can be rescued upon detergent removal. Authors should discuss this or at least highlight the primary caveat of the peptidisc method of membrane protein reconstitution - which is that it begins with detergent solubilization of the proteome and does not completely circumvent structural perturbations invoked by detergents.

We would like to clarify that, in our current workflow, ligand incubation occurs after reconstitution into Peptidiscs. As such, the method is designed to circumvent the negative effects of detergent during the critical steps involving low-affinity interactions.

That said, we fully acknowledge that Peptidisc reconstitution begins with detergent solubilization (e.g., 1% DDM), and we have revised the conclusion to explicitly state this important caveat. As the reviewer correctly points out, this initial step may introduce some structural perturbations or result in the loss of weakly associated lipid modulators.

However, reconstitution into Peptidiscs rapidly restores a detergent-free environment for membrane proteins, which has been shown in our previous studies [PMID: 38577106, 38232390, 31736482, 31364989] to mitigate these effects. Specifically, we have demonstrated that time-limited DDM exposure, followed by Peptidisc reconstitution, minimizes membrane protein delipidation, enhances thermal stability, retains functionality, and preserves multi-protein assemblies.

It would also be important to test detergents that are even milder than 1% DDM and ones which are harsher than 1% DDM to show that this method of reconstitution can indeed rescue the perturbations to the structure and interactions of the membrane protein done by detergents during solubilization step.

We selected 1% DDM based on our previous work [PMID: 37295717, 39313981, 38232390], where it consistently enabled robust and reproducible solubilization for Peptidisc reconstitution. We agree that comparing milder detergents (e.g., LMNG) and harsher ones (e.g., SDC) would provide valuable insights into how detergent strength influences structural perturbations, and how effectively these can be mitigated by Peptidisc reconstitution. Preliminary data (not shown) from mouse liver membranes indicate broadly similar proteomic profiles following solubilization with DDM, LMNG, and SDC, although potential differences in functional activity or ligand binding remain to be investigated.

Based on the methods provided, it appears that the final amount of detergent in peptidisc membrane protein library was 0.008%, which is ~150 μ M. The CMC of DDM depending on the amount of NaCl could be between 120-170 μ M.

While we cannot entirely rule out the presence of residual DDM (0.008%) in the raw library, its free concentration may be lower than initially estimated. This is related to the formation of mixed micelles with the amphipathic peptide scaffold, which is supplied in excess during reconstitution. These mixed micelles are subsequently removed during the ultrafiltration step. Furthermore, in related work using His-tagged Peptidiscs [PMID: 32364744], we purified the library by nickel-affinity chromatography following a 5 $\times$ dilution into a detergent-free buffer. Although this purification step reduced the number of soluble proteins, the same membrane proteins were retained, suggesting that any residual detergent does not significantly interfere with Peptidisc reconstitution. Supporting this, our MM-TPP assays on purified libraries (data not shown) consistently demonstrated stabilization of ATP-binding proteins (e.g., SLC27A2, DNAJB3), indicating that the observed ligand-protein interactions result from successful incorporation into Peptidiscs.

Perhaps, to completely circumvent the perturbations from detergents other methods of detergentfree solubilization such as using SMA polymers and SMALP reconstitution could

be explored for a comparison. Moreover, a comparison of the peptidisc reconstitution with detergent-free extraction strategies, such as SMA copolymers, could lend more strength to the presented method.

We agree that detergent-free methods such as SMA polymers hold promise for membrane protein solubilization. However, in preliminary single-replicate experiments using SMA2000 at 51 °C in the presence of ATP-VO₄ (data not shown), we observed broad, non-specific stabilization effects. Of the 2,287 quantified proteins, 9.3% were annotated as ATP-binding, yet 9.9% of the 101 proteins showing a log₂ fold change >1 or <-1 were ATPbinding, indicating no meaningful enrichment. Given this lack of specificity and the limited dataset, we chose not to pursue further SMA experiments and have not included them here. However, in a recent study (<https://doi.org/10.1101/2025.08.25.672181>), we directly compared Peptidisc, SMA, and nanodiscs for liver membrane proteome profiling. In that work, Peptidisc outperformed both SMA and nanodiscs in detecting membrane protein dysregulation between healthy and diseased liver. By extension, we expect Peptidisc to offer superior sensitivity and specificity for detecting ligand-induced stabilization events, such as those observed here with ATP-vanadate.

Cross-verification of the identified interactions, and subsequent stabilization or destabilizations, should be demonstrated by other in vitro methods of thermal stability and ligand binding analysis using purified protein to support the efficacy of the MM-TPP method. An example cross-verification using SDS-PAGE, of the well-studied MsbA, is shown in Figure 2. In a similar fashion, other discussed targets such as, BCS1L, P2RX4, DgkA, Mao-B, and some un-annotated IMPs shown in supplementary figure 3 that display substantial stabilization or destabilization should be cross-verified.

We appreciate this suggestion and note that a similar point was raised in R1's comment "In addition, except for MsbA, no orthogonal methods are used to support the conclusions, and the authors rely entirely on quantifying rather small differences in abundances using either iBAQ or LFQ." We have developed a detailed response to R1 on this matter, which equally applies here.

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Reviewer #1 (Recommendations for the authors):

"The authors use iBAC or LFQ to compare across samples. This inconsistency is puzzling. As far as I know, LFQ should always be used when comparing across samples"

As mentioned above, we use iBAQ only in Fig. 2B to illustrate within-sample relative abundance; all comparative analyses elsewhere use LFQ. We have updated the Fig. 2B legend to state this explicitly.

We used iBAQ Fig. 2B as it provides a notion of protein abundance within a sample, normalizing the summed peptide intensities by the number of theoretically observable peptides. This normalization facilitates comparisons between proteins within the same sample, offering a clearer understanding of their relative molar proportions [PMID: 33452728]. LFQ, by contrast, is optimized for comparing the same protein across different samples. It achieves this by performing delayed normalization to reduce run-to-run variability and by applying maximal peptide ratio extraction, which integrates pairwise peptide intensity ratios across all samples to build a consistent protein-level quantification matrix [PMID: 24942700]. These features make LFQ more robust to missing values and technical variation, thereby enabling accurate detection of relative abundance changes in the same protein under different experimental conditions. This distinction is well supported by the proteomics literature: Smits et al. [PMID: 23066101] used iBAQ specifically to determine the relative abundance of proteins within one sample, whereas LFQ was applied for comparative analyses between conditions.

"[Regarding Figure 2A] Why does the control also contain ATP-vanadate? Also, I am not aware of a commercially available chemical "ATP-VO₄". I assume this is a mistake"

The control condition in Figure 2A was mislabeled, and the figure has been corrected to remove this discrepancy. In our experiments, ATP and orthovanadate (VO₄) were added together, and for simplicity this was annotated as "ATP-VO₄."

"[Regarding Figure 2B] What is the fold change in MsbA iBAQ values? It seems that the differences are quite small, and as such require a more quantitative approach than iBAQ (e.g. SILAC or some other internal standard). In addition, what information does this panel add relative to 2C"

The figure has been updated to clarify that the values shown are log₂transformed iBAQ intensities. Figures 2B and 2C are complementary: Figure 2B shows that in the control sample, MsbA's peptide abundance decreases with temperatures (51, 56, and 61 °C) relative to the remaining bulk proteins. Figure 2C shows the specific thermal profiles of MsbA in control and ATP-vanadate conditions. To make this clearer, we have added a sentence to the Results section explaining the specific role of Figure 2B.

Together, these panels indicate that the method can identify ligand-induced stabilization even for proteins whose abundance decreases faster than the bulk during the TPP assay. We have provided the rationale for not using SILAC or TMT labeling in our public response.

"[Regarding Figure 2C] Although not mentioned in the legend, I assume this is iBAQ quantification, which as mentioned above isn't accurate enough for such small differences. In addition, I find this data confusing: why is MsbA more stable at the lower temperatures in the absence of ATP-vanadate? The smoothed-line representation is misleading, certainly given the low number of data points"

The data presented represent LFQ values for MsbA, and we have updated the figure legend to clearly indicate this. Additionally, as suggested, we have removed the smoothing line to more accurately reflect the data. Regarding the reviewer's concern about stability at lower temperatures, we note that MsbA exhibits comparable abundance at 38 °C and 46 °C under both conditions, with overlapping error bars. We therefore interpret these data as indicating no significant difference in stability at the lower temperatures, with ligand-dependent stabilization becoming apparent only at elevated temperatures. We do not exclude the possibility that MsbA stability at these temperatures is affected by the conformational dynamics of this ABC transporter upon ATP binding and hydrolysis.

"[Regarding Figure 3A] is this raw LFQ data? Why did the authors suddenly change from iBAQ to LFQ? I find this inconsistency puzzling"

To clarify, all analyses of protein stabilization or destabilization presented in the manuscript are based on LFQ values. The only instance where iBAQ was used is Figure 2B, where it served to illustrate the relative peptide abundance of MsbA within the same sample. We have revised the figure legends and text to make this distinction explicit and ensure consistency in presentation.

"[Regarding Figure 3B] The non-specific ATP-dependent stabilization increases the likelihood of false positive hits. This limitation is not mentioned by the authors. I think it is important to show other small molecules, in addition to ATP. The authors suggest that their approach is highly relevant for drug screening. Therefore, a good choice is to test an effect of a known stabilizing drug (eg VX-809 and CFTR)"

We thank the reviewer for this suggestion. As noted in the manuscript (results and discussion sections), ATP is a natural hydrotrope and is therefore expected to induce broad, non-specific stabilization effects, a phenomenon also observed in previous proteome-wide studies, which demonstrated ATP's widespread influence on cytosolic protein solubility and thermal stability (PMID: 30858367). To demonstrate that MM-TPP can resolve specific ligand–protein interactions beyond these global ATP effects, we tested 2-methylthio-ADP (2-MeS-ADP), a selective agonist of P2RY12 (PMID: 14755328). In these experiments, we observed robust and reproducible stabilization of P2RY12 at both 51°C and 57°C, with no consistent stabilization of unrelated proteins across temperatures. This provides direct evidence that our workflow can distinguish specific from non-specific ligand-induced effects. We selected 2-MeS-ADP due to its structural stability and receptor higher-affinity over ADP, allowing us to extend our existing workflow while testing a receptor-specific interaction. We agree that extending this approach to clinically relevant small-molecule drugs, such as VX-809 with CFTR, would further underscore the pharmacological potential of MM-TPP, and we have now noted this as an important avenue for future studies.

"X axis of Figure 3B: Log 2 fold difference of what? iBAQ? LFQ? Similar ambiguity regarding the Y axis of 3E. What peptide? And why the constant changes in estimating abundances?"

We thank the reviewer for pointing out these inaccuracies in the figure annotations. As mentioned above, all analyses (except Figure 2B) are based on LFQ values. We have revised the figure legends and text to make this clear.

In Figure 3E, "peptide intensity" refers to log₂ LFQ peptide intensities derived from the BCS1L protein, as indicated in the figure caption.

"The authors suggest that P2RY6 and P2RY12 are stabilized by ADP, the hydrolysis product of ATP. Currently, the support for this suggestion is highly indirect. To support this claim, the authors need to directly show the effect of ADP. In reference to the alpha fold results shown in Figure 4D, the authors state that "Collectively, these data highlight the ability of MM-TPP to detect the side effects of parent compounds, an important consideration for drug development". To support this claim, it is necessary to show that Mao-B is indeed best stabilized with ADP or AMP, rather than ATP."

In this revision, we chose not to test ADP directly, as it is a broadly binding, relatively weak ligand that would likely stabilize many proteins without revealing clear target-specific effects. Since we had already evaluated ATP-VO₄, a similarly broad, non-specific ligand, additional

testing with ADP would provide limited additional insight. Instead, we prioritized 2-methylthio-ADP, a selective agonist of P2RY12, to more effectively demonstrate the specificity of MM-TPP. With this ligand, we observed clear and reproducible stabilization of P2RY12, underscoring the ability of MM-TPP to resolve receptor–ligand interactions beyond ATP’s broad hydrotropic effects. Importantly, and as expected, we did not observe stabilization of the related purinergic receptor P2RY6, further supporting the specificity of the observed effect.

We have also revised the AlphaFold-related statement in Figure 4D to adopt a more cautious tone: “Collectively, these data suggest that MM-TPP may detect potential side effects of parent compounds, an important consideration for drug development.” In this context, we use AlphaFold not as a validation tool, but rather as a structural aid to help rationalize why certain off-target proteins (e.g., ATP with Mao-B) exhibit stabilization.

Reviewer #2 (Recommendations for the authors):

“In the main text, it will be useful to include the unique peptides table of at least the targets discussed in the manuscript. For example, in presence of AMP-PNP at 51oC P2RY6 shows 4-6 peptides in all n=3 positive & negative ionization modes. But, for P2RY12 only 1-3 peptides were observed. Depending on the sequence length and the relative abundance in the cell of a protein of interest, the number of peptides observed could vary a lot per protein. Given the unique peptide abundance reported in the supplementary file, for various proteins in different conditions, it appears the threshold of observation of two unique peptides for a protein to be analyzed seems less stringent.”

By applying a filter requiring at least two unique peptides in at least one replicate, we exclude, on average, 15–20% of the total identified proteins. We consider this a reasonable level of stringency that balances confidence in protein identification with the retention of relevant data. This threshold was selected because it aligns with established LC-MS/MS data analysis practices (PMID: 32591519, 33188197, 26524241), and we have included these references in the Methods section to justify our approach. We have included in this revision a Supplemental Table 2 showing the unique peptide counts for proteins highlighted in this study.

“It appears that the time of heat treatment for peptidisc library subjected to MM-TPP profiling was chosen as 3 min based on the results presented in Supplementary Figure 1A, especially the loss of MsbA observed in 1% DDM after 3 min heat perturbation. However, when reconstituted in peptidisc there seems to be no loss in MsbA even after 12 mins at 45oC. So, perhaps a longer heat treatment would be a more efficient perturbation.”

Previous studies indicate that heat exposure of 3–5 minutes is optimal for visualizing protein denaturation (PMID: 23828940, 32133759). We have added a statement to the Results section to justify our choice of heat exposure. Although MsbA remains stable at 45 °C for extended periods, higher temperatures allow for more effective perturbation to reveal destabilization. Supplementary Figure 1A specifically illustrates MsbA instability in detergent environments.

“Some of the stabilized temperatures listed in Table 1 are a bit confusing. For example, ABCC3 and ABCG2. In the case of ABCC3 stabilization was observed at 51oC and 60oC, but 56oC is not mentioned. In the same way, 51oC is not mentioned for ABCG2. You would expect protein to be stabilized at 56oC if it is stabilized at both 51oC and 60oC. So, it is unclear if the stabilizations were not monitored for these proteins at the missing temperatures in the table or if no peptides could be recorded at these temperatures as in the case of P2RX4 at 60oC in Figure 4C.”

Both scenarios are represented in our data. For some proteins, like ABCG2, sufficient peptide coverage was achieved, but no stabilization was observed at intermediate temperatures (e.g., 56 °C), likely because the perturbation was not strong enough to reveal an effect. In other cases, such as ABCC3 at 56 °C or P2RX4 at 60 °C, the proteins were not detected due to insufficient peptide identifications at those temperatures, which explains their omission from the table.

"In Figure 4C, it is perplexing to note that despite $n = 3$ there were no peptide fragments detected for P2RX4 at 60°C in presence of ATP-VO₄, but they were detected in presence of AMP-PNP. It will be useful to learn authors explanation for this, especially because both of these ligands destabilize P2RX4. In Figure 4B, it would have been great to see the effect of ADP too, to corroborate the theory that ATP metabolites could impact the thermal stability."

In Figure 4C, the absence of P2RX4 peptide detection at 60 °C with ATP-VO₄ mirrors variability observed in the corresponding control ($n = 6$). Specifically, neither the control nor ATP-VO₄ produced unique peptides for P2RX4 at 60 °C in that replicate, whereas peptides were detected at 60 °C in other replicates for both the control and AMP-PNP, and at 64 °C for ATP-VO₄, the controls, and AMP-PNP. Such missing values are a natural feature of MS-based proteomics and can arise from multiple technical factors, including inconsistent heating, incomplete digestion, stochastic MS injection, or interference from Peptidisc peptides. We therefore interpret the absence of peptides in this replicate as a technical artifact rather than evidence against protein destabilization. Importantly, the overall dataset consistently shows that both ATP-VO₄ and AMP-PNP destabilize P2RX4, supporting their characterization as broad, non-specific ligands with off-target effects.

Because ATP and ADP belong to the same class of broadly binding, non-specific ligands, additional testing with ADP would not provide meaningful mechanistic insight. Instead, we chose to test 2-methylthio-ADP, a selective P2RY12 agonist. This experiment revealed robust, reproducible stabilization of P2RY12, without consistent effects on unrelated proteins at 51 °C and 57 °C, thereby demonstrating the ability of MM-TPP to detect specific receptor–ligand interactions.

Finally, we note that P2RX4 is not a primary target of ATP-VO₄ or AMP-PNP. Consequently, the observed destabilization of P2RX4 is expected to be less pronounced than the strong, physiologically consistent stabilization of ABC transporters by ATP-VO₄, as shown in Figure 3D, where the majority of ABC transporters are thermally stabilized across all tested temperatures.

"As per Figure 4, P2Y receptors P2RY6 and P2RY12 both showed great thermal stability in presence of ATP-VO₄ despite their preference for ADP. The authors argue this could be because of ATP metabolism, and binding of the resultant ADP to the P2RY6. If P2RX4 prefers ATP and not the metabolized product ADP that apparently is available, ideally you should not see a change in stability. A stark destabilization would indicate interaction of some sorts. P2X receptors are activated by ATP and are not naturally activated by AMP-PNP. So, destabilization of P2RX4 upon binding to ATP that can activate P2X receptors is conceivable. However, destabilization both in presence of ATP-VO₄ and AMP-PNP is unclear. It is perhaps useful to test effect of ADP using this method, and maybe even compare some antagonists such as TNPATP."

In this study, we did not directly test ADP, as we had already demonstrated that MM-TPP detects stabilization by broad-binding ligands such as ATP-VO₄. Instead, we focused on a more selective ligand, 2-MeS-ADP, a specific agonist of P2RY12 [PMID: 14755328]. Here, we observed robust and reproducible stabilization of P2RY12 at 51 °C and 57 °C, while P2RY6

showed no significant changes, and no other proteins were consistently stabilized (Figure 4B, S4). This confirms that MM-TPP can distinguish specific ligand–receptor interactions from broader ATP-induced effects. To further explore the assay’s nuance and sensitivity, testing additional nucleotide ligands—including antagonists like TNP-ATP or ATP γ S—would provide valuable insights, and we have identified this as an important future direction.

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